European Commission



SPECIAL EUROBAROMETER 554

Artificial Intelligence and the future of work

EUROBAROMETER REPORT SUMMARY

FIELDWORK: April – May 2024

FACE TO FACE EUROBAROMETER

This survey has been requested by the European Commission, Directorate-General for Employment, Social Affairs and Inclusion (DGEMPL)

and co-ordinated by the European Commission, Directorate-General for Communication (DGCOMM 'Media monitoring and Eurobarometer' Unit)

This document does not represent the point of view of the European Commission. The interpretations and opinions contained in it are solely those of the authors.

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Introduction

Artificial intelligence, or AI, refers to computer systems capable of performing tasks that typically require human intelligence. As AI develops, it can offer benefits to citizens and businesses across Europe, and it can lead to increase in productivity and job creation. However, the rise of AI also gives rise to challenges. The stock of industrial robots in Europe has more than quadrupled in the past 25 years. Technological advancement generates increasing uncertainty. Technology may have a significant impact on the quantity of jobs available to humans, but it also transforms them, changing how jobs are performed, with implications for workers' quality of life and for productivity.

The European Commission recognised this potential early and is taking action to foster the take-up of artificial intelligence by the private and public sectors. In its Communication of April 2018 on Artificial Intelligence for Europe, the Commission outlines a number of measures and financing instruments through which it will promote this goal[1]. Moreover, linking to the European Digital Strategy[2], in July 2020, the European Commission launched the European Skills Agenda for sustainable competitiveness, social fairness and resilience, which sets ambitious objectives for upskilling (improving existing skills) and reskilling (training in new skills). Since then, much progress has been made in each of the 12 flagship actions identified[3], including the publication in early 2022 of the updated Digital Competence Framework covering the digital competences needed for emerging technologies and AI[4].

The accelerated digitalisation of workplaces also puts the spotlight on issues related to surveillance, the use of data, and the application of algorithmic management tools. Algorithms can be used in management functions, such as planning, organisation, command, coordination and control. The Joint Research Centre's AMPWork study finds that a relatively high proportions of workers in Spain and Germany use digital tools and are subject to digital monitoring and algorithmic management[5]. AI systems are also often applied to guide recruitment, monitor workloads, or define remuneration rates.

Addressing the challenges of algorithmic decision-making, notably the risks of biased decisions, discrimination and lack of transparency, can improve trust in AI-powered systems, promote their uptake and protect fundamental rights[6]. Proposed by the European Commission in December 2021 and provisionally agreed upon by the European Parliament and the Council in February 2024, the Platform Work Directive will be the first EU instrument to tackle the challenges arising from the use of AI in the workplace. Once finally adopted, it will provide more transparency, fairness, human oversight, safety and accountability regarding algorithmic management on digital labour platforms by helping people better understand how tasks are allocated and prices are set. It will also enable them to contest decisions that affect working conditions if needed.

Against this backdrop, this Special Eurobarometer survey aims to gauge EU citizens' perceptions of and attitudes towards automation and AI in working life. More particularly, the study covers the following topics:

- General perceptions of the impact of digital technologies;
- Proficiency with digital technologies;
- Awareness of and experience with digital technologies, including AI, in the workplace;
- Attitudes towards these digital technologies and their use in the workplace.

The survey finds that, while positive perceptions of the impact of the most recent digital technologies have declined over the past seven years, Europeans still generally have a positive view of said impact. Specifically, while citizens generally see the positive potential of technologies and specifically artificial intelligence, this does not translate to unfettered support. Notably, more than six in ten Europeans say they think robots and artificial intelligence area good thing for society, with the caveat that a clear majority (84%) also say such technologies need careful management. The survey highlights that the potential of robots and artificial intelligence to lighten workload—by doing boring or repetitive jobs or being used to make accurate decisions—is appreciated, but respondents are wary of their impact on social areas like communication between colleagues.

Looking specifically at digital skills and awareness, most Europeans think of their current skills as sufficient for their daily life and work needs. However, this result should be seen in context: a significant number of Europeans lack even basic digital skills (almost 45%)[7] and the number of ICT specialists is growing but still relatively low (est. 9.8

[1] 'Communication Artificial Intelligence for Europe', COM(2018) 237 final: https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:52018DC0237.
[2] 'Shaping Europe's digital future': https://digital-strategy.ec.europa.eu/en/policies.
[3] 'Factsheet – European Skills Agenda: progress on the 11 flagship actions (2023)'. Available at: https://ec.europa.eu/social/main.jsp?catId=1223.
[4] 'Digital Competences Framework (DigComp 2.2) update published': https://ec.europa.eu/social/main.jsp?langId=en&catId=89&newsId=10193&furtherNews=yes.
[5] 'Algorithmic management and digital monitoring of work': https://joint-research-centre.ec.europa.eu/scientific-activities-z/employment/algorithmic-management-and-digital-monitoring-work_en.
[6] 'The European Pillar of Social Rights Action Plan': https://op.europa.eu/webpub/empl/european-pillar-of-social-rights/en.
[7] 'Digital skills in 2023: impact of education and age': https://ec.europa.eu/eurostat/web/products-eurostat-news/w/ddn-20240222-1

million)[8]. Most employees are also aware of their employer's use of digital technologies to manage employee activities.

Patterns in the opinions of Europeans based on their socio-demographic characteristics are also highlighted throughout the report. Younger respondents, respondents with less financial difficulties, respondents who are currently in employment or those with a higher education level tend to have more positive attitudes towards technologies and artificial intelligence, for example.

Some of the questions in this Special Eurobarometer were asked in previous Eurobarometer surveys conducted in March 2017[9] and in December 2019[10]. Where possible, the results of the current survey are compared with those of the two previous studies.

[8] 'More people employed in ICT in the EU in 2023': https://ec.europa.eu/eurostat/web/products-eurostat-news/w/ddn-20240524-2

[9] Special Eurobarometer 460, 'Attitudes towards the impact of digitisation and automation on daily life': https://europa.eu/eurobarometer/surveys/detail/2160.

[10] Special Eurobarometer 503, 'Attitudes towards the impact of digitalisation on daily lives': https://europa.eu/eurobarometer/surveys/detail/2228.

Methodology

This survey was carried out by the Verian Group in the 27 EU Member States between the 25th of April and the 22nd of May 2024. Some 26,415 respondents from different social and demographic groups were interviewed in their mother tongue. This survey was commissioned by the European Commission, Directorate-General for Employment, Social Affairs and Inclusion (DG EMPL).

Note: In this report, Member States are referred to by their official abbreviation, as listed below:

Belgium	BE	Lithuania	LT
Bulgaria	BG	Luxembourg	LU
Czechia	CZ	Hungary	HU
Denmark	DK	Malta	MT
Germany	DE	Netherlands	NL
Estonia	EE	Austria	AT
Ireland	IE	Poland	PL
Greece	EL	Portugal	PT
Spain	ES	Romania	RO
France	EN	Slovenia	SI
Croatia	HR	Slovakia	SK
Italy	IT	Finland	FI
Republic of Cyprus*	CY*	Sweden	SE
Latvia	LV		

European Union - weighted average for the 27 Member States of the European Union	EU27
BE, FR, IT, LU, DE, AT, ES, PT, IE, NL, FI, EL, EE, SI, CY, MT, SK, LV, LT, HR	euro area
BG, CZ, DK, HU, PL, RO, SE	Non-euro area

* Cyprus as a whole is one of the 27 European Union Member States. However, the "acquis communautaire" has been suspended in the part of the country which is not controlled by the government of the Republic of Cyprus. For practical reasons, only the interviews carried out in the part of the country controlled by the government of the Republic of Cyprus are included in the "CY" category and in the EU27 average.

We wish to thank the people throughout the European Union who have given their time to take part in this survey.
Without their active participation, this study would not have been possible.



Key findings

At least half of Europeans believe the most recent digital technologies, including Artificial Intelligence, have a positive impact on each of the areas tested

- Two thirds of respondents currently in employment think the most recent digital technologies, including Artificial Intelligence, currently have a positive impact on their job (66%).
- Of all the respondents, more than six in ten believe the most recent digital technologies currently have a positive impact on the economy, their quality of life and social security benefits (all 62%).
- More than half (56%) indicate that the most recent digital technologies have a positive impact on society.

Large majorities think that rules addressing risks and maximising the benefits of digital technologies in the workplace would be important

- More than eight in ten (82%) say protecting workers' privacy would be important in addressing risks and maximizing the benefits of digital technologies, including Artificial Intelligence, in the workplace.
- Similarly, more than seven in ten think it would be important to involve workers and their representatives in the design and adoption of new technologies (77%), to enforce more transparency in the use of digital technologies to handle HR decision-making (75%), to prohibit fully automated decision-making processes (74%) and to limit the automated monitoring of employees (72%).

Majorities consider themselves to be sufficiently skilled in the use of the most recent digital technologies

- Around seven in ten consider themselves to be sufficiently skilled in the use of the most recent digital technologies, including Artificial Intelligence, to be able to benefit from digital and online learning opportunities (71%, +14 percentage points since March 2017) and in their daily life (70%). The respondents who answered 'Not applicable' are not included in the analysis of this question in order to ensure the trend comparability.
- Among those currently in employment and who did not answer 'Not applicable' at the aforementioned question, three quarters feel they have a sufficient level of digital skills to do their job (75%, -7 pp).
- More than seven in ten respondents who are not retired and did not answer 'Not applicable' believe they possess an adequate digital proficiency to do a future job if they were to find a job or to change jobs within the next twelve months (72%, +9 pp).

Most Europeans have a positive opinion of robots and Artificial Intelligence in the workplace, but concerns about their negative impact on employment are still present

- More than six in ten (62%, +1 pp) positively perceive the use of robots and Artificial Intelligence in the workplace, while close to one third (32%, +2 pp) have a negative perception. These perceptions however vary notably by country, with 86% of respondents in Denmark saying they perceive this positively compared to less than half in Portugal or Greece (both 48%).
- Majorities agree that robots and Artificial Intelligence increase the pace at which workers complete tasks (73%), that Artificial Intelligence is necessary as it can do jobs that are seen as boring or repetitive (66%) and that robots and Artificial Intelligence can be used to make accurate decisions in the workplace (53%).
- At the same time, 66% believe that due to the use of robots and Artificial Intelligence, more jobs will disappear than new jobs will be created, and that robots and Artificial Intelligence steal peoples' jobs. However, both these proportions have declined since March 2017 (-8 pp and -6 pp, respectively).
- In addition, 61% agree that robots and Artificial Intelligence have a negative impact on communication between colleagues.

Large majorities think robots and Artificial Intelligence are a good thing for society, but also that they should be carefully managed

- More than six in ten (63%) think that robots and Artificial Intelligence help people do their jobs or carry out daily tasks at home. This proportion shows a decrease of five percentage points since March 2017.
- Nonetheless, more than eight in ten (84%, -4 pp) think that these technologies require careful management.
- Less than half (48%) indicate that robots and Artificial Intelligence should be used more widely outside the workplace, while 43% hold the opposite view.

The use of digital technologies, including Artificial Intelligence, for many activities in the workplace is seen in a negative light

- Overall, respondents are concerned about the use of digital technologies, including Artificial Intelligence, to automatically fire workers (78%), to monitor workers (63%), to assess workers' performance (57%), to select applicants for a job (57%), to gather additional information on applicants for a job (50%), and to collect, process, and store workers' personal data (49%).
- However, around two thirds (67%) positively perceive the use of such technologies to improve workers' safety and security, and a relative majority positively perceive their use to allocate tasks to workers or manage their working schedules and shifts (49% 'positively' vs 44% 'negatively').

A majority is aware of the use their employer makes of digital technologies and sizeable proportions report that digital technologies have been used to perform activities in their workplace

- More than six in ten of those who are currently in employment (62%) say they are aware of the use their employer makes of digital technologies, including Artificial Intelligence, to manage their or their co-workers' activities.
- Among those who have a current or past occupation, 30% report that digital technologies, including Artificial Intelligence, have enforced safety measures in their current or previous workplaces, and 29% say these technologies have managed worktime schedules.
- More than one in five indicate that monitoring workers' activities (24%), allocating tasks to workers (22%) or assessing workers' performance, including imposing sanctions or attributing rewards (21%) have been performed by these technologies in their workplaces. In addition, 18% indicate that these technologies have been used to hire workers.

Less than half of those currently in employment say their employer informed them about the use of digital technologies, while more than half of employers and managers report having done so

- Among respondents who are currently in employment, close to half (49%) say their employer informed them about the use of digital technologies, including Artificial Intelligence, to manage activities in their workplace.
- This proportion includes 16% who say they have been made aware of this, but without further details, and 18% who report having received a detailed explanation, including information about the benefits, drawbacks, and their rights.

A large majority of those currently in employment feels their employer equips them with the necessary tools to work with digital technologies

- More than two thirds of those currently in employment (68%) agree that their employer provides them with the necessary tools or training to work effectively with the most recent digital technologies, while 28% disagree with the statement.
- The level of agreement with this statement is higher among those aged less than 55, those with a higher education level, managers and other white collars, and those working in larger establishments.



I. General perceptions of digital technologies

1. Impact of digital technologies

Majorities think digital technologies have a positive impact on most areas tested[11], even though there is a decline in positive views since 2017

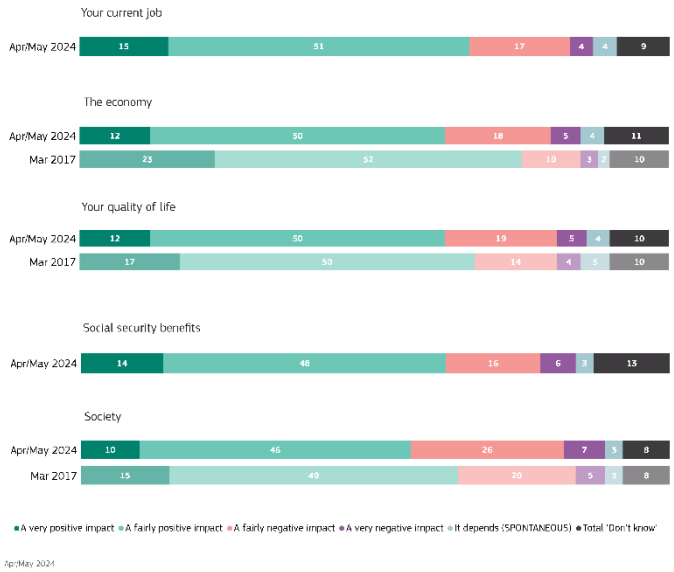
Two thirds of respondents who are currently working think the most recent digital technologies, including Artificial Intelligence[12], currently have a positive impact on their job (66%), compared to one in five who say they have a negative impact.

More than six in ten of total respondents think the most recent digital technologies, including Artificial Intelligence, currently have a positive impact on the economy, their quality of life and social security benefits[13] (all 62%), while more than two in ten (between 22% and 24%) say these technologies have a negative impact on these areas.

More than half (56%) consider that the most recent digital technologies have a positive impact on society, while 33% say these have a negative impact.

Compared to the last time this question was asked in March 2017, there have been declines in the shares of respondents who think the most recent technologies have a positive impact on the economy (-13 percentage points), society (-8 pp) and their quality of life (-5 pp)[14].

QB1. In your view, what impact do the most recent digital technologies, including Artificial Intelligence, currently have on (EU27) (%)



[11] QB1. In your view, what impact do the most recent digital technologies, including Artificial Intelligence, currently have on... 1) The economy; 2) Society; 3) Your quality of life; 4) Your current job; 5) Social security benefits. A very positive impact; A fairly positive impact; A fairly negative impact; A very negative impact; Don't know enough about the most recent digital technologies (SPONTANEOUS); It depends (SPONTANEOUS); Don't know.

[12] The next questions are about digital technologies, including Artificial Intelligence. Digital technologies refer to technologies that use data and computing to help people do their work more efficiently. They include hardware such as smartphones, laptops or robots, and software, such as smartphone applications, video conferencing tools, collaboration platforms, or project management tools in the workplace. Artificial Intelligence refers to computer systems capable of performing tasks that typically require human intelligence. It is used, for instance, in driverless cars or drones, in healthcare to improve medical diagnoses, and in various other applications such as answering questions and offering support to users on websites or call centres. Artificial Intelligence can also be used in workplaces, for example to monitor and evaluate workers or automatically assign them certain tasks. Artificial Intelligence can also be used to create new content such as images, text, or music as well as to answer questions and assist with tasks such as composing emails or essays. ChatGPT is the best-known example'.

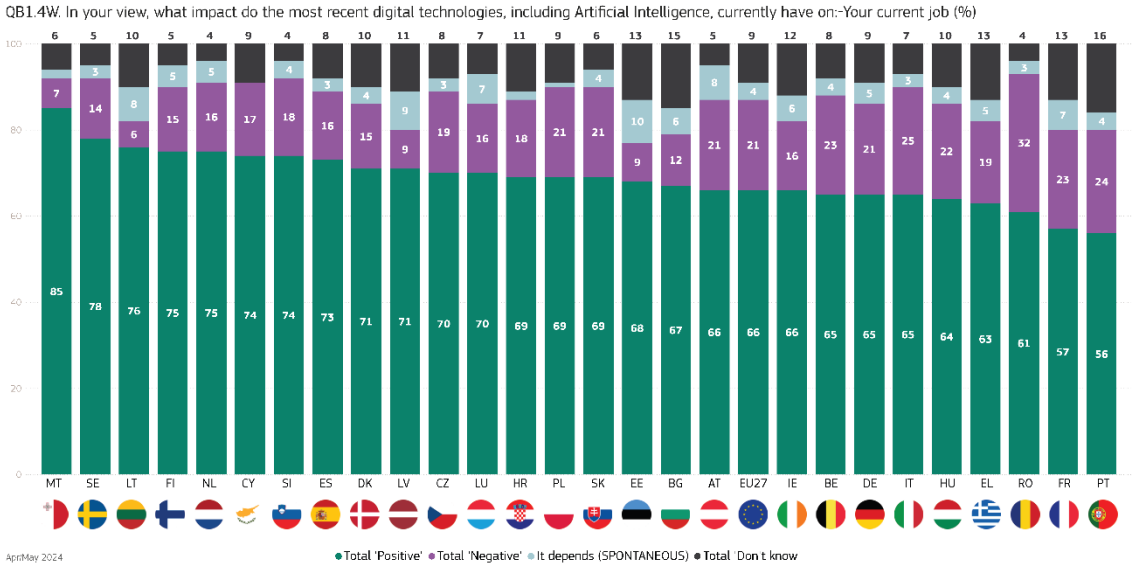
[13] Respondents were provided with the following explanation of the use of the most recent digital technologies for 'social security benefits': 'For social security benefits we mean accessing information online on people's rights and obligations; claiming healthcare reimbursements, unemployment benefits, maternity benefits or pensions through a phone app or website; having benefits automatically granted, adjusted or removed based on one's life events'.

[14] It is important to note that the introduction and the phrase 'including Artificial Intelligence' have been added to the wording of this question compared to the 2017 survey.

a) Current employment

The proportions of respondents who think the impact of the most recent technologies on their current job is positive vary widely across EU Member States. This answer is given by a majority in all EU countries, particularly in Malta (85%), Sweden (78%) and Lithuania (76%).

At the opposite end of the scale, 56% in Portugal, 57% in France and 61% in Romania have a positive opinion.

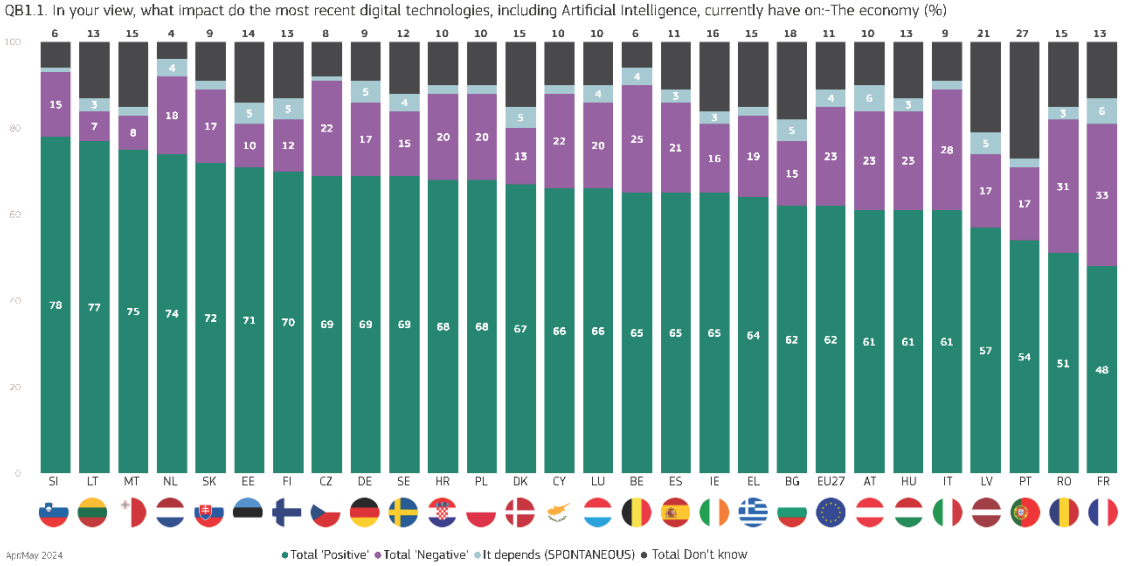


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b) The economy

At country level, the majority in 26 out of the 27 EU Member States believes the most recent technologies, including Artificial Intelligence, currently have a positive impact on the economy, with this share being the highest in Slovenia (78%), Lithuania (77%) and Malta (75%).

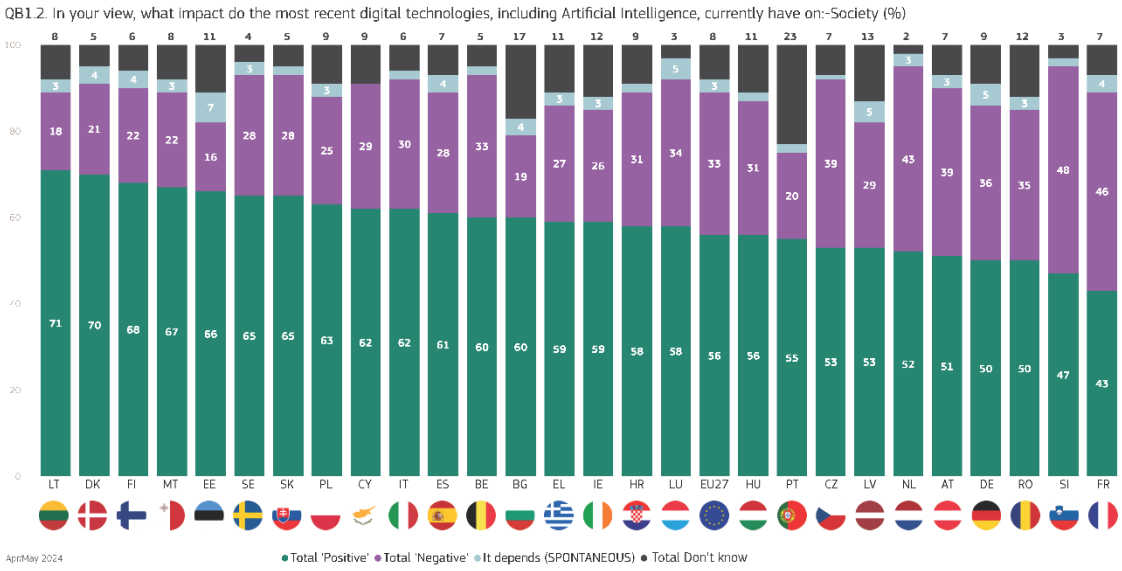
At the other end of the spectrum, this answer is given by a minority in France (48%) and by 51% in Romania and 54% in Portugal.



c) Society

In 23 countries, more than half think that the most recent technologies have a positive impact on society, most notably in Lithuania (71%), Denmark (70%) and Finland (68%).

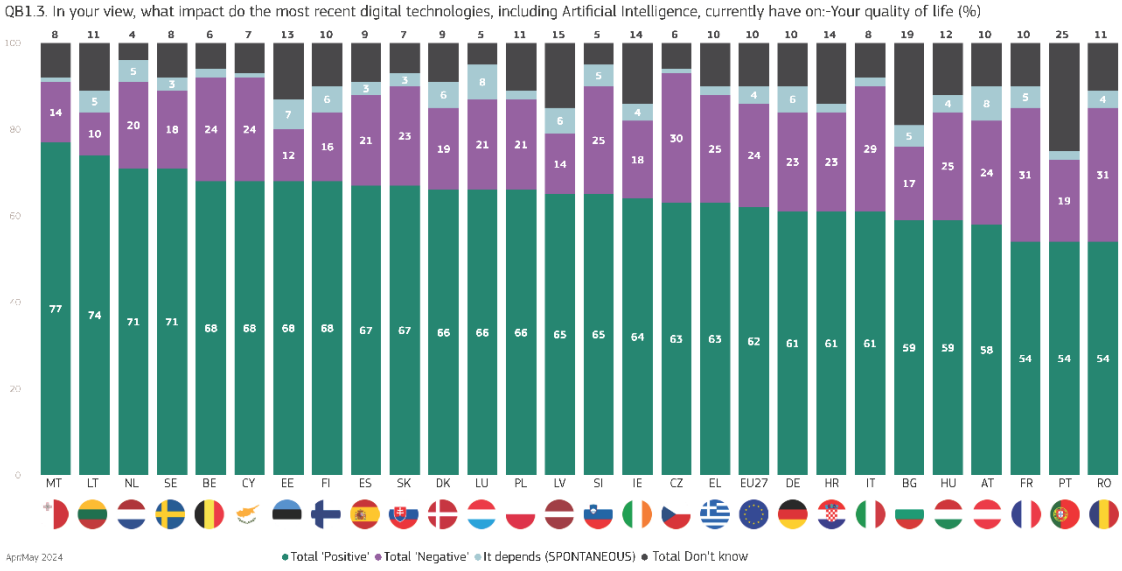
This compares to half of the respondents or less who give this answer in France (43%), Slovenia (47%) and Germany and Romania (both 50%).



d) Quality of life

The country-level analysis shows that a majority in all 27 EU Member States believes that the most recent digital technologies currently have a positive impact on their quality of life.

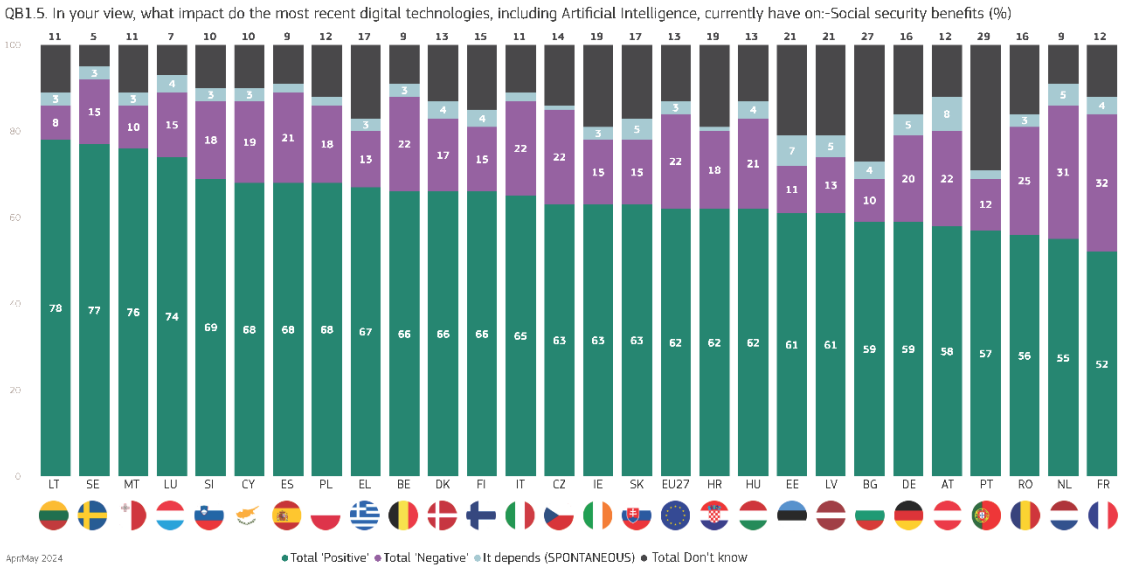
This proportion ranges from more than seven in ten in Malta (77%), Lithuania (74%) and the Netherlands and Sweden (both 71%) to 54% in France, Portugal and Romania.

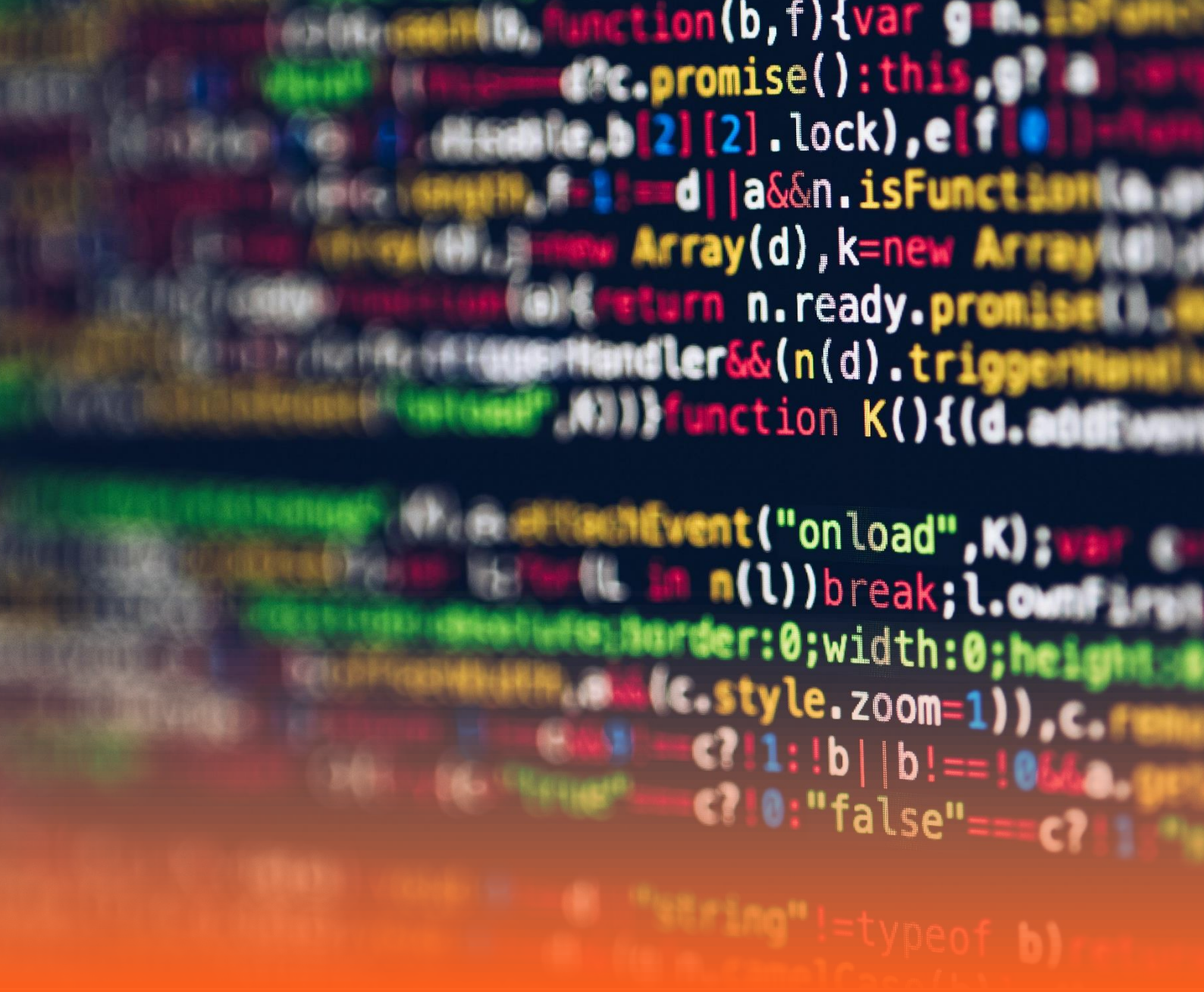


e) Social security benefits

At country level, majorities in all 27 EU Member States believe the impact of the most recent technologies on social security benefits is currently positive, with more than three quarters giving this answer in Lithuania (78%), Sweden (77%) and Malta (76%).

This compares to 52% in France, 55% in the Netherlands and 56% in Romania who give this answer.





II. Proficiency with digital technologies

1. Digital skills overall

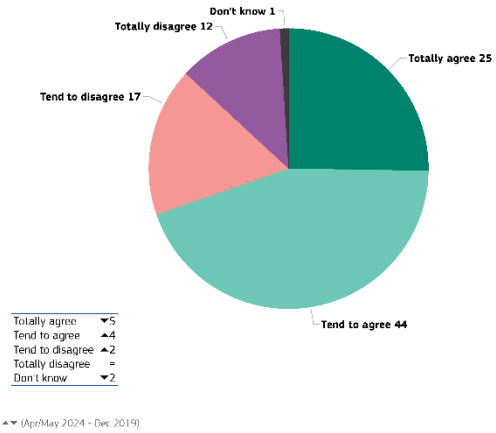
a) In daily life

Most Europeans consider themselves to be sufficiently skilled in the use of the most recent digital technologies in their daily life[15]

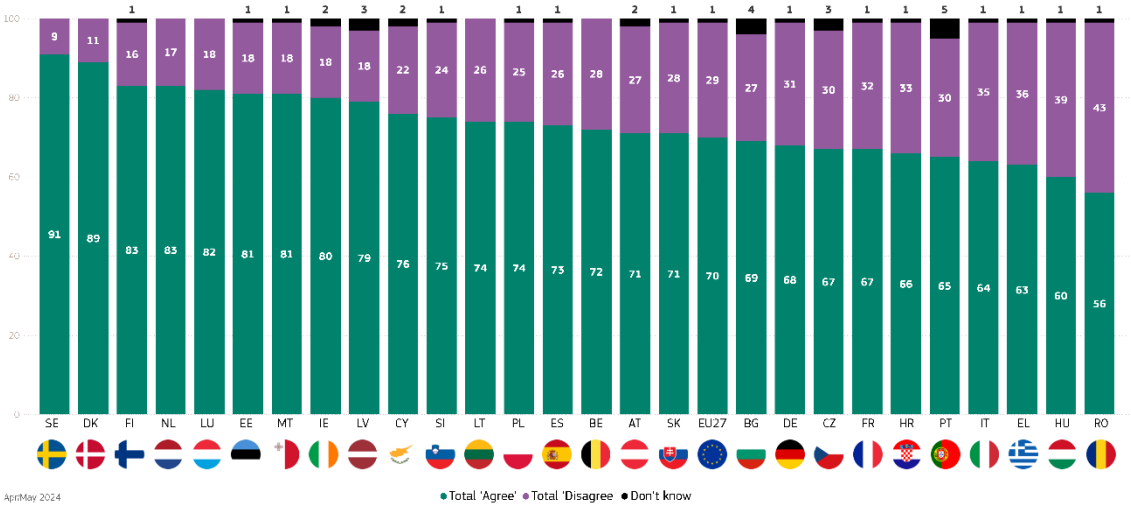
Seven in ten respondents (70%) consider themselves to be sufficiently skilled in the use of the most recent digital technologies, including Artificial Intelligence, in their daily life, with 25% who 'totally agree' with the statement, showing a decrease of 5 percentage points since March 2017 Close to three in ten (29%, +2 pp) do not consider themselves to be sufficiently skilled, including 12% who are in total disagreement with the statement, while 1% say they don't know (-3 pp).

In all 27 EU Member States, a majority of respondents consider themselves to be sufficiently skilled in the use of the most recent digital technologies, including Artificial Intelligence, in their daily life. Around nine in ten give this answer in Sweden (91%) and Denmark (89%), and more than eight in ten do so in Finland and the Netherlands (both 83%). This compares to more than half who think they are sufficiently skilled in Romania (56%) and at least six in ten in Hungary (60%) and Greece (63%).

QB2.1. To what extent do you agree or disagree with the following statements regarding your skills in the use of the most recent digital technologies, including Artificial Intelligence? Think about, for example, using smart appliances at home that you can control remotely for heating, cooking, or cleaning; using apps on your smartphone to handle your daily schedule; or using websites to manage your tasks at work. You consider yourself to be sufficiently skilled in the use of digital technologies...:-... in your daily life (EU27) (%)



QB2.1. To what extent do you agree or disagree with the following statements regarding your skills in the use of the most recent digital technologies, including Artificial Intelligence? Think about, for example, using smart appliances at home that you can control remotely for heating, cooking, or cleaning; using apps on your smartphone to handle your daily schedule; or using websites to manage your tasks at work. You consider yourself to be sufficiently skilled in the use of digital technologies...:-... in your daily life (%)



[15] QB2. To what extent do you agree or disagree with the following statements regarding your skills in the use of the most recent digital technologies, including Artificial Intelligence? Think about, for example, using smart appliances at home that you can control remotely for heating, cooking, or cleaning; using apps on your smartphone to handle your daily schedule; or using websites to manage your tasks at work. You consider yourself to be sufficiently skilled in the use of digital technologies... 1) ...in your daily life; 2) ...to do your job; 3) ... to do a future job if you were to find a job or to change jobs within the next twelve months; 4) ...to be able to benefit from digital and online learning opportunities. Totally agree; Tend to agree; Tend to disagree; Totally disagree; Not applicable (SPONTANEOUS); Don't know.

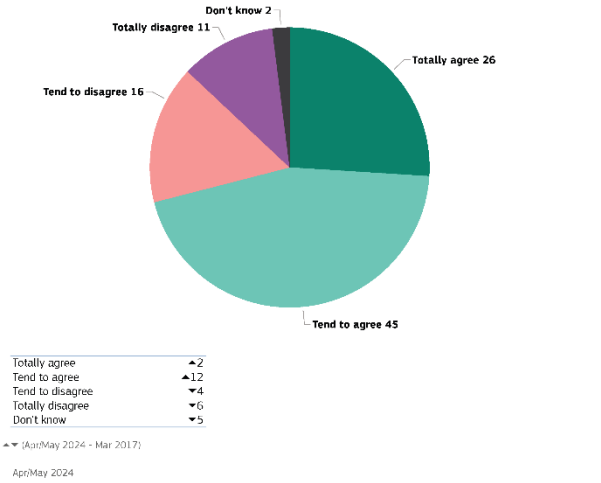
b) Digital and learning opportunities

Around seven in ten think they possess sufficient digital skills to be able to benefit from digital and online learning opportunities

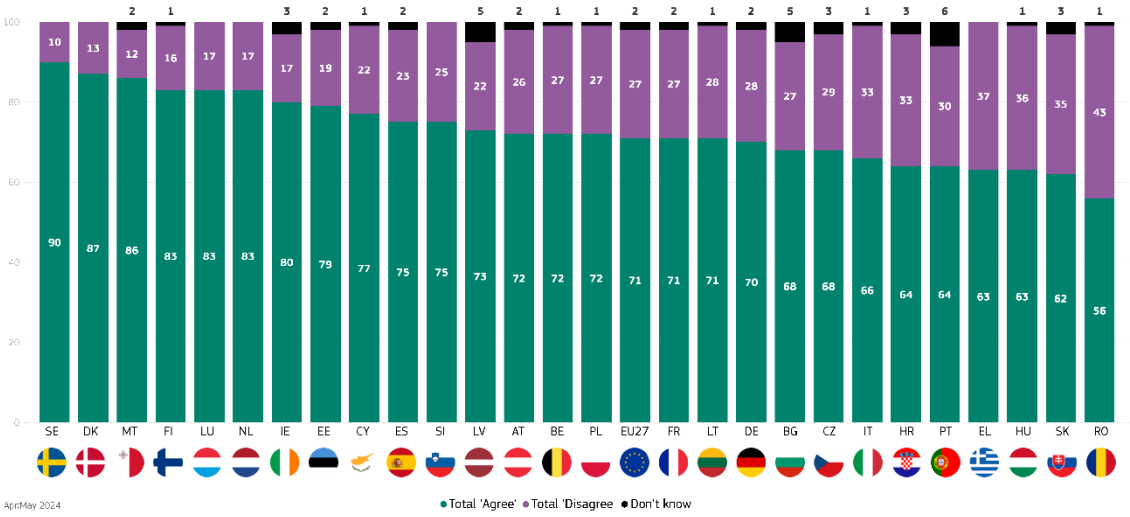
Slightly more than seven in ten respondents (71%, +14 pp) see themselves as sufficiently skilled in the use of the most recent digital technologies, including Artificial Intelligence, to be able to benefit from digital and online learning opportunities. This proportion includes 26% who 'totally agree' this is the case (+2 pp). By contrast, 27% do not consider themselves sufficiently skilled (-10 pp), with more than one in ten (11%, -6 pp) who 'totally disagree' with the statement, while 1% say they don't know[16].

In 18 out of 27 EU Member States at least seven in ten consider themselves to be sufficiently skilled in the use of the most recent digital technologies, including Artificial Intelligence, to be able to benefit from digital and online learning opportunities. As was the case with digital skills in daily life, respondents in Sweden (90%), Denmark (87%), are the most likely to be confident about the use of digital technologies to benefit from learning opportunities, followed by Malta (86%). At the opposite end of the scale, once again, more than half of those in Romania (56%) agree with the statement, followed by Slovakia (64%) and Hungary and Greece (both 63%).

QB2.4. To what extent do you agree or disagree with the following statements regarding your skills in the use of the most recent digital technologies, including Artificial Intelligence? Think about, for example, using smart appliances at home that you can control remotely for heating, cooking, or cleaning; using apps on your smartphone to handle your daily schedule; or using websites to manage your tasks at work. You consider yourself to be sufficiently skilled in the use of digital technologies...:-... to be able to benefit from digital and online learning opportunities (EU27) (%)



QB2.4. To what extent do you agree or disagree with the following statements regarding your skills in the use of the most recent digital technologies, including Artificial Intelligence? Think about, for example, using smart appliances at home that you can control remotely for heating, cooking, or cleaning; using apps on your smartphone to handle your daily schedule; or using websites to manage your tasks at work. You consider yourself to be sufficiently skilled in the use of digital technologies...:-... to be able to benefit from digital and online learning opportunities (%)



[16] The answer 'Not applicable' was introduced in the latest survey wave. In order to preserve trend comparability, this answer was removed from the analysis of QB2 and therefore the base size does not reflect the total sample.

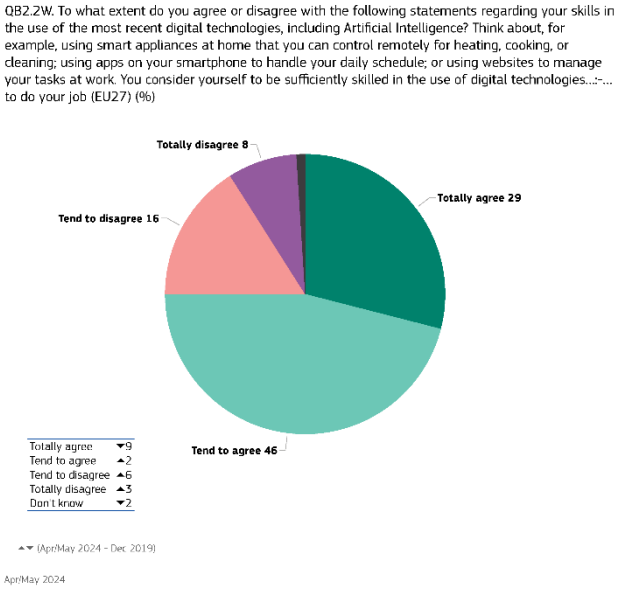
2. Digital skills and labour market

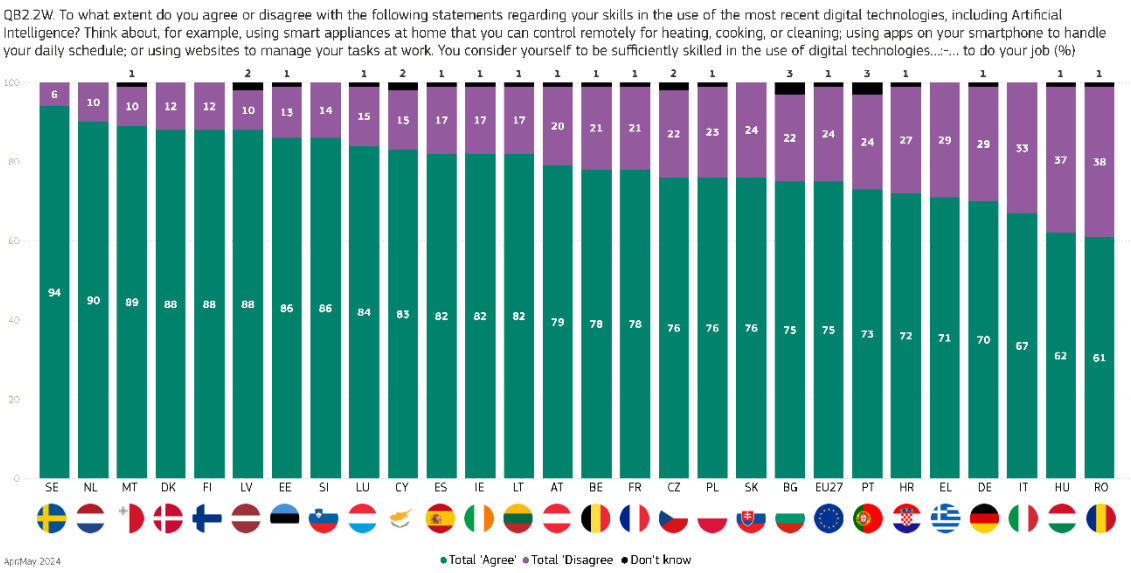
a) Current employment

The majority of Europeans are confident about their level of digital skills to do their job

Three quarters of respondents who are currently working (75%, -7 pp) consider themselves to be sufficiently skilled in the use of the most recent digital technologies, including Artificial Intelligence, to do their job, including 29% who 'totally agree' this is the case (-9 pp). Conversely, close to one quarter (24%, +9 pp) do not consider themselves to be sufficiently skilled, while 1% say they don't know (-2 pp). The answer 'Not applicable' is not included in the result in order to preserve the trend comparability[17].

In all 27 EU Member States, at least six in ten respondents agree that they are sufficiently skilled in the use of the most recent digital technologies, including Artificial Intelligence, to do their job. This proportion ranges from more than nine in ten in Sweden (94% -6 pp) and around nine in ten in the Netherlands (90%, -3 pp), and Malta (89%, +16) to more than six in ten in Romania (61%, -7 pp) and Hungary (62%, -11 pp) and over two thirds in Italy (67%, -4 pp).





[17] The answer 'Not applicable' was introduced in the latest survey wave. In order to preserve trend comparability, this answer was removed from the analysis of QB2 and therefore the base size does not reflect the total sample.

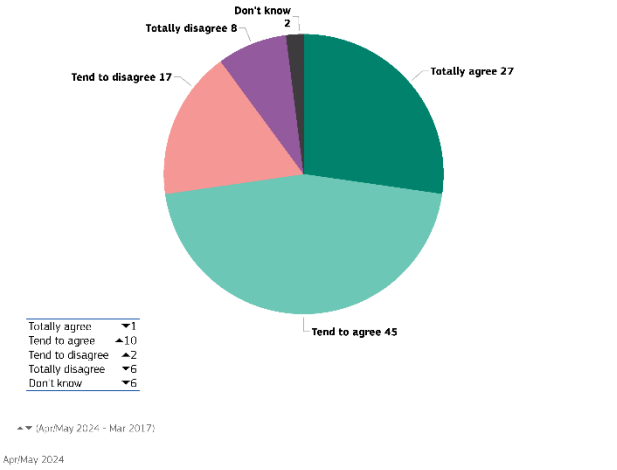
b) Future employment

Most European see their digital skills as sufficient to do a future job or to change jobs within the next twelve months

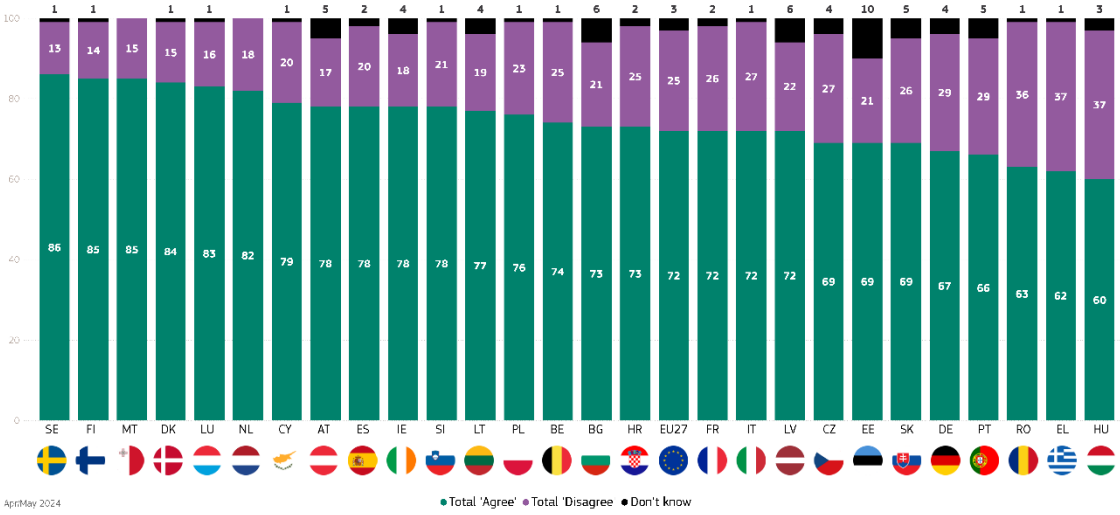
Among those who are not retired, more than seven in ten (72%, +9 pp) agree that they are sufficiently skilled in the use of the most recent digital technologies, including Artificial Intelligence, to do a future job if they were to find a job or to change jobs within the next twelve months. This share of respondents includes more than one quarter (27%, -1 pp) who are in total agreement with the statement. By contrast, one quarter (25%, -2 pp) do not see themselves as sufficiently skilled, while 3% say they don't know (-5 pp)[18].

In 19 countries, at least seven in ten respondents are confident about their level of digital skills to do a future job if they were to find a job or to change jobs within the next twelve months. This proportion is highest in Sweden (87, +20 pp), Finland (85%, +23) and Malta (85%, +21 pp), while it is lowest in Hungary (60%, +13 pp) and Greece (63%, +16 pp) and in Romania (63%, +7 pp).

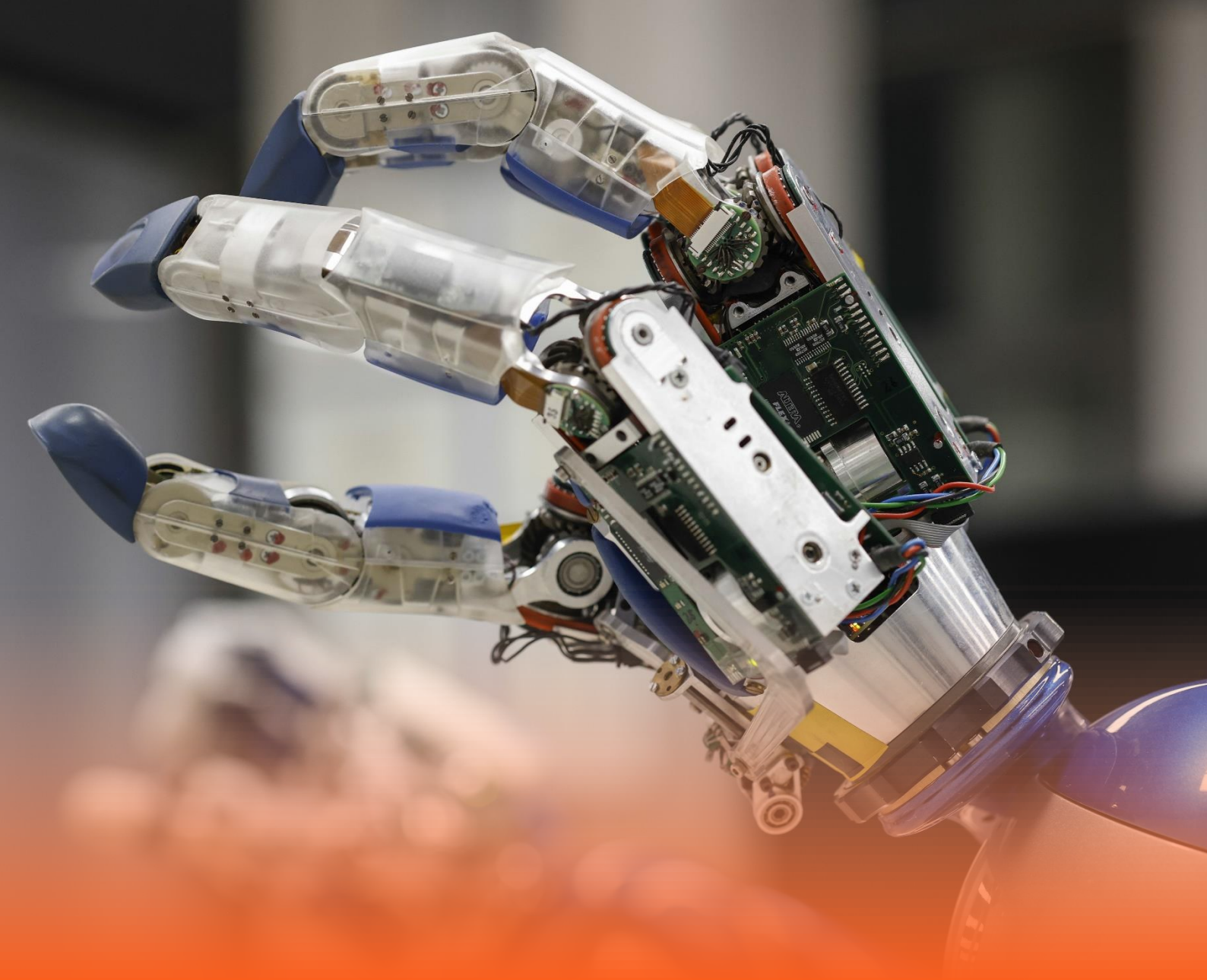
QB2.3W. To what extent do you agree or disagree with the following statements regarding your skills in the use of the most recent digital technologies, including Artificial Intelligence? Think about, for example, using smart appliances at home that you can control remotely for heating, cooking, or cleaning; using apps on your smartphone to handle your daily schedule; or using websites to manage your tasks at work. You consider yourself to be sufficiently skilled in the use of digital technologies...:-... to do a future job if you were to find a job or to change jobs within the next twelve months (EU27) (%)



QB2.3W. To what extent do you agree or disagree with the following statements regarding your skills in the use of the most recent digital technologies, including Artificial Intelligence? Think about, for example, using smart appliances at home that you can control remotely for heating, cooking, or cleaning; using apps on your smartphone to handle your daily schedule; or using websites to manage your tasks at work. You consider yourself to be sufficiently skilled in the use of digital technologies...:-... to do a future job if you were to find a job or to change jobs within the next twelve months (%)



[18] The answer 'Not applicable' was introduced in the latest survey wave. In order to preserve trend comparability, this answer was removed from the analysis of QB2 and therefore the base size does not reflect the total sample.



III. Attitudes towards digital technologies and Artificial Intelligence in the workplace

1. Perception of robots and Artificial Intelligence

a) In the workplace

The majority of Europeans have a positive view of the use of robots and AI in the workplace

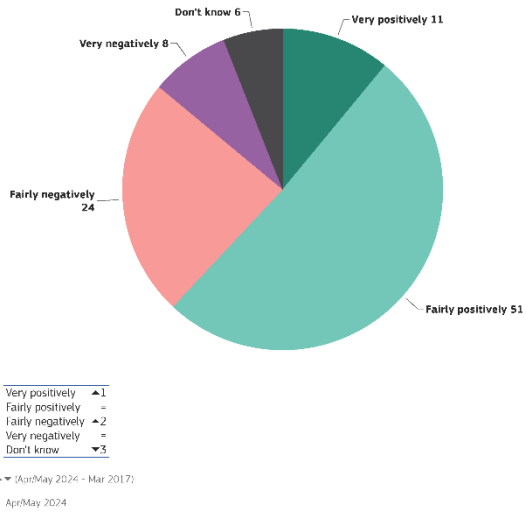
More than six in ten (62%, +1 percentage point since March 2017) positively perceive the use of robots and Artificial Intelligence in the workplace, including around one in ten (11%, +1 pp) who perceive this 'very positively'[19]. By contrast close to one third (32%, +2 pp) have a negative perception and around one in twenty (6%, -3 pp) say they don't know.

Perceptions of the use of robots and Artificial Intelligence in the workplace have remained broadly stable since March 2017[20].

In 25 out of the 27 EU Member States, a majority of respondents positively perceives the use of robots and Artificial Intelligence in the workplace, most notably in Denmark (86%, +4 pp), Finland (80%, +9 pp) and Sweden (79%, -1 pp). Conversely, a minority of respondents hold a positive view in Greece and Portugal (both 48%, -1 pp; and -7 pp) and 53% are of this opinion in France (-2 pp).

In 17 countries, respondents are more likely than they were in March 2017 to positively perceive the use of robots and Artificial Intelligence in the workplace. This is most notably the case of respondents in Slovakia (+10 percentage points) and in Finland and Malta (both +9 pp).

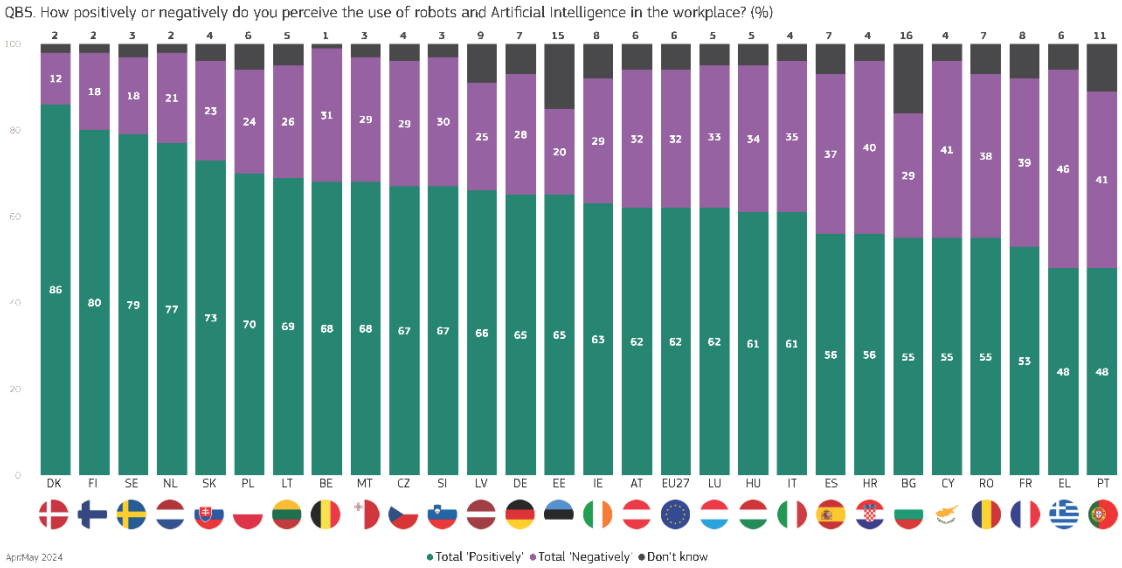
QB5. How positively or negatively do you perceive the use of robots and Artificial Intelligence in the workplace? (EU27) (%)



Conversely, this share of respondents has decreased in eight countries, particularly in Bulgaria and Estonia (both -8 percentage points) and in Portugal (-7 pp).

This figure has remained unchanged in Italy and Spain

QB5. How positively or negatively do you perceive the use of robots and Artificial Intelligence in the workplace? (%)



[19] QB5. How positively or negatively do you perceive the use of robots and Artificial Intelligence in the workplace? By robots we mean machines designed to perform tasks which are normally performed by humans. They can perform them by themselves or, often, working together with a human. For example, robots that can perform surgery together with a surgeon or automatic tills in supermarkets. Very positively; Fairly positively; Fairly negatively; Very negatively; Don't know.

[20] It should be noted that, compared to the 2017 survey, the definition of 'robots' was changed.

b) In the labour market

Majorities agree with statements on the benefits of robots and Artificial Intelligence, but also think these will **lead to job destruction and workers' replacement**

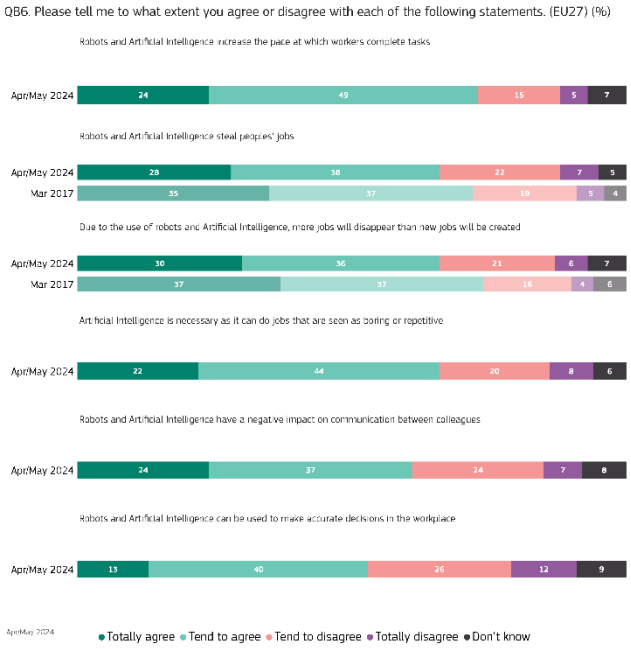
Respondents were asked whether they agree or disagree with a series of statements regarding the impact of robots and Artificial Intelligence on the labour market and on working life[21].

Close to three quarters of the respondents (73%) agree that robots and Artificial Intelligence increase the pace at which workers complete tasks, including around one quarter (24%) who 'totally agree'. Conversely, one in five disagree with this statement.

Moreover, two thirds (66%) agree that Artificial Intelligence is necessary as it can do jobs that are seen as boring or repetitive, including 22% who 'totally agree', while close to three in ten (28%) disagree with this statement.

The same level of agreement can be found for the statement 'due to the use of robots and Artificial Intelligence, more jobs will disappear than new jobs will be created', with three in ten who find themselves in total agreement (66%). The total agreement level has decreased by eight percentage points since March 2017. This affects the share of respondents who 'totally agree', which has also decreased (-7 pp). By contrast, more than one quarter (27%, +7 pp) disagree that more jobs will disappear than new jobs will be created.

Similarly, 66% agree that robots and Artificial Intelligence steal peoples' jobs, with nearly three in ten (28%) who are in total agreement. The overall agreement level has declined since March 2017 (-6 percentage points) due to a decrease in the share of those who 'totally agree' (-7 pp). Conversely, just below three in ten (29%, +5 pp) are in disagreement with this statement.



Around six in ten (61%) agree that robots and Artificial Intelligence have a negative impact on communication between colleagues, with close to one quarter (24%) who are in total agreement. This compares to around three in ten (31%) who disagree that this is the case.

Finally, more than half (53%) indicate that robots and Artificial Intelligence can be used to make accurate decisions in the workplace, including 13% who 'totally agree'. Nearly four in ten (38%) disagree with this statement, with 12% being in total disagreement.

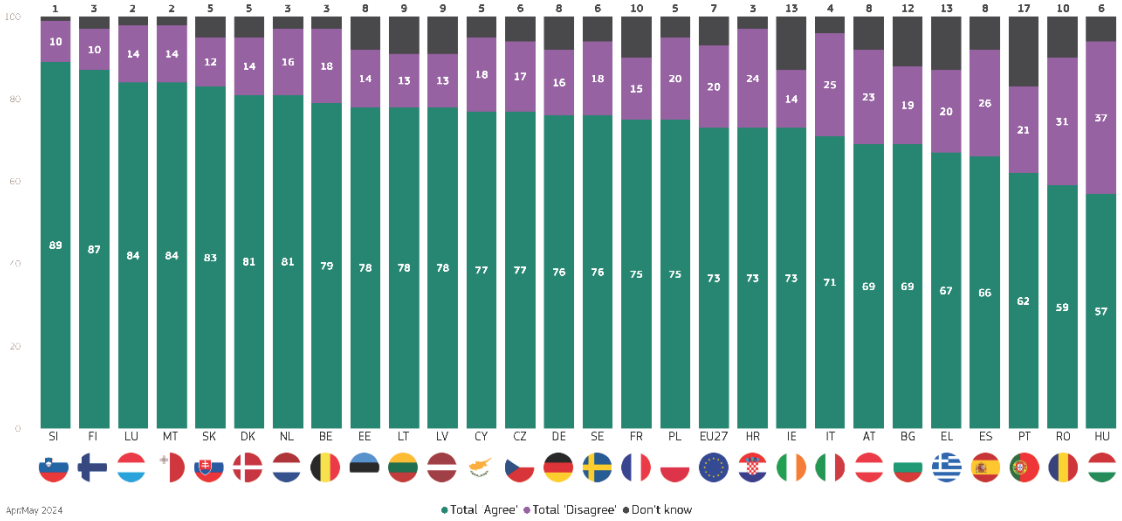
For each of these statements, less than one in ten respondents say they don't know (between 5% and 9%)

[21] QB6. Please tell me to what extent you agree or disagree with each of the following statements. 1) Due to the use of robots and Artificial Intelligence, more jobs will disappear than new jobs will be created; 4) Artificial Intelligence is necessary as it can do jobs that are seen as boring or repetitive; 5) Robots and Artificial Intelligence steal peoples' jobs; 6) Robots and Artificial Intelligence increase the pace at which workers complete tasks; 7) Robots and Artificial Intelligence have a negative impact on communication between colleagues; 8) Robots and Artificial Intelligence can be used to make accurate decisions in the workplace. Totally agree; Tend to agree; Tend to disagree; Totally disagree; Don't know.

In 17 EU Member States, at least three quarters agree that robots and Artificial Intelligence increase the pace at which workers complete tasks, most notably in Slovenia (89%) and Finland (87%) and in Luxembourg and Malta (both 84%).

At the other side of the spectrum, 57% in Hungary, 59% in Romania and 62% in Portugal are in agreement.

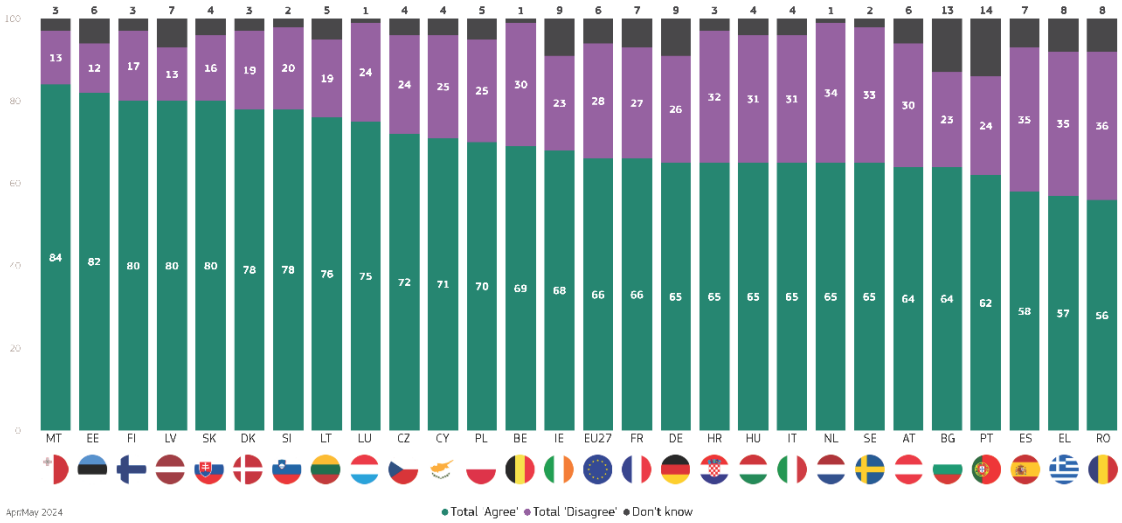
QB6.6. Please tell me to what extent you agree or disagree with each of the following statements.:-Robots and Artificial Intelligence increase the pace at which workers complete tasks (%)



In 14 countries, more than two thirds agree that Artificial Intelligence is necessary as it can do jobs that are seen as boring or repetitive.

This proportion ranges from at least eight in ten in Malta (84%) and Estonia (82%) and in Finland, Latvia and Slovakia (all 80%) to less than six in ten in Romania (56%), Greece (57%) and Spain (58%).

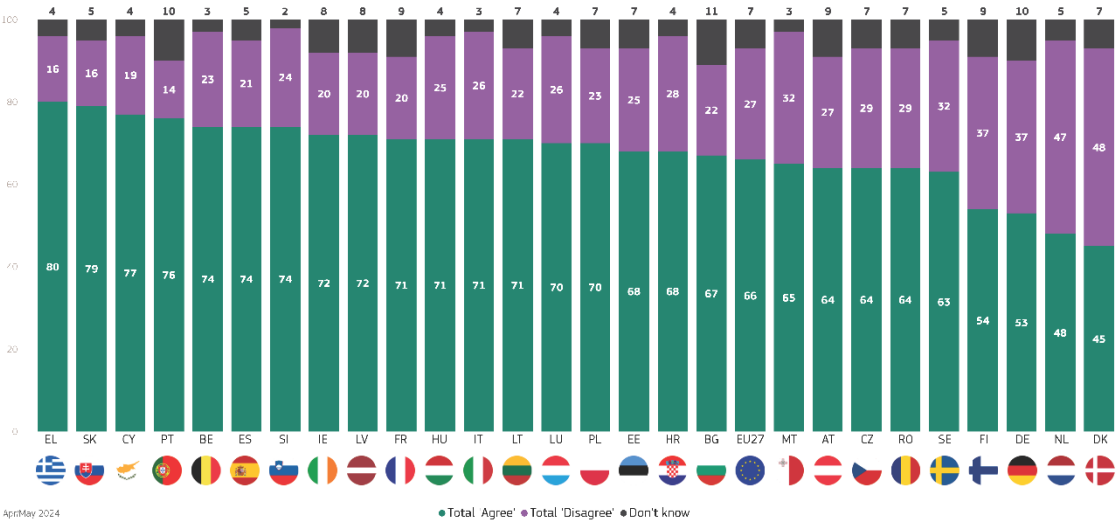
QB6.4. Please tell me to what extent you agree or disagree with each of the following statements.:-Artificial Intelligence is necessary as it can do jobs that are seen as boring or repetitive (%)



In 15 countries, at least seven in ten agree that, due to the use of robots and Artificial Intelligence, more jobs will disappear than new jobs will be created.

Agreement levels are highest in Greece (80%), Slovakia (79%) and Cyprus (77%), while they are lowest in Denmark (45%), the Netherlands (48%) and Germany (53%).

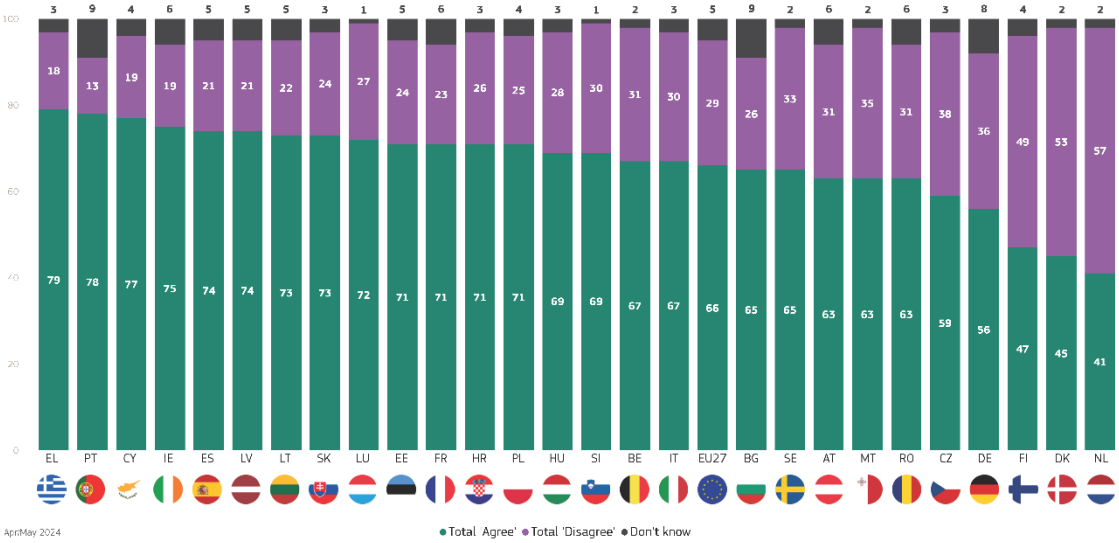
QB6.1. Please tell me to what extent you agree or disagree with each of the following statements.:-Due to the use of robots and Artificial Intelligence, more jobs will disappear than new jobs will be created (%)



More than two thirds of the respondents in 17 countries agree that robots and Artificial Intelligence steal **peoples' jobs**, with more than three quarters who give this answer in Greece (79%), Portugal (78%) and Cyprus (77%).

At the opposite end of the scale, a minority of respondents agree in the Netherlands (41%), Denmark (45%) and Finland (47%).

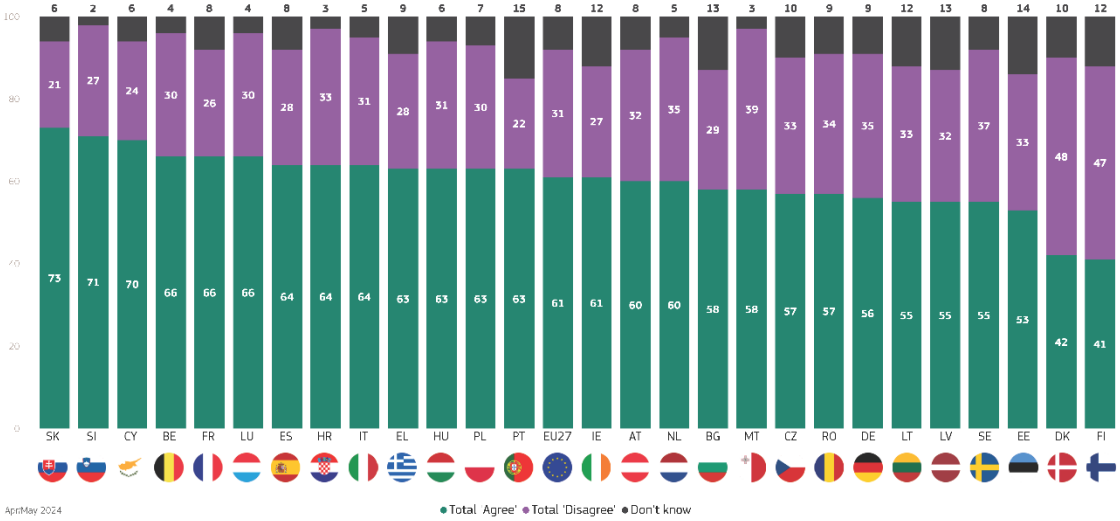
QB6.5. Please tell me to what extent you agree or disagree with each of the following statements.:-Robots and Artificial Intelligence steal peoples' jobs (%)



In 16 EU Member States, at least six in ten agree that robots and Artificial Intelligence have a negative impact on communication between colleagues, with agreement levels being the highest in Slovakia (73%), Slovenia (71%) and Cyprus (70%).

Minorities agree with this statement in Finland (41%) and Denmark (42%), as do 53% in Estonia.

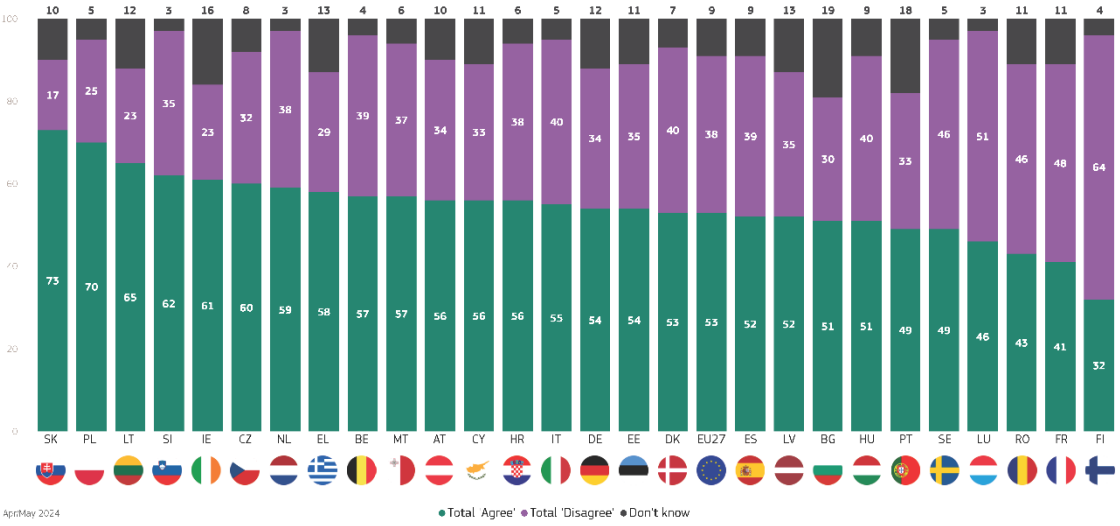
QB6.7. Please tell me to what extent you agree or disagree with each of the following statements:-Robots and Artificial Intelligence have a negative impact on communication between colleagues (%)



In 21 EU Member States, the majority agrees that robots and Artificial Intelligence can be used to make accurate decisions in the workplace. Respondents in Slovakia (73%), Poland (70%) and Lithuania (65%) are the most likely to hold this view.

While those in Finland (32%) as well as those in France (41%) and Romania (43%) are the least likely to do so.

QB6.8. Please tell me to what extent you agree or disagree with each of the following statements:-Robots and Artificial Intelligence can be used to make accurate decisions in the workplace (%)



2. Perceived impact of robots and Artificial Intelligence

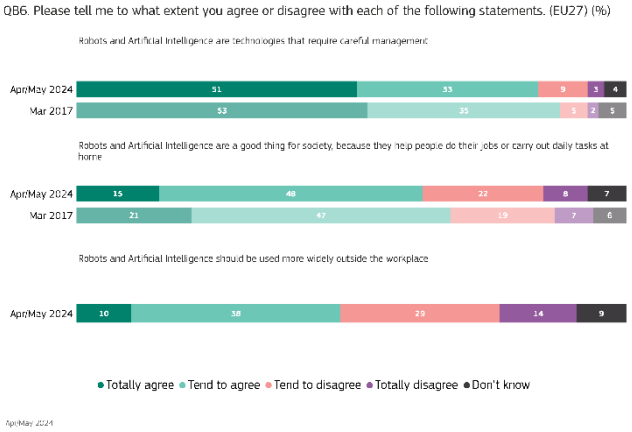
a) Overall

Most respondents agree robots and Artificial Intelligence are a good thing for society, but a majority think they should be carefully managed and a minority agree they should be used more widely[22]

More than eight in ten (84%) agree that robots and Artificial Intelligence are technologies that require careful management, with a majority of respondents (51%) in total agreement. The level of agreement with this statement has decreased by four percentage points compared to March 2017. By contrast, just above one in ten (12%, +5 pp) disagree that these technologies require careful management.

In spite of this, more than six in ten (63%) still think that robots and Artificial Intelligence are a good thing for society, because they help people do their jobs or carry out daily tasks at home, including 15% who are in total agreement with this statement. The overall agreement level has declined since March 2017 (-5 percentage points) due to a decrease in the share of respondents who are in total agreement (-6 pp).

However, less than half (48%) believe that robots and Artificial Intelligence should be used more widely outside the workplace, including one in ten who 'totally agree' with this statement.



Conversely, more than four in ten (43%) hold the opposite view, with 14% who are in total disagreement.

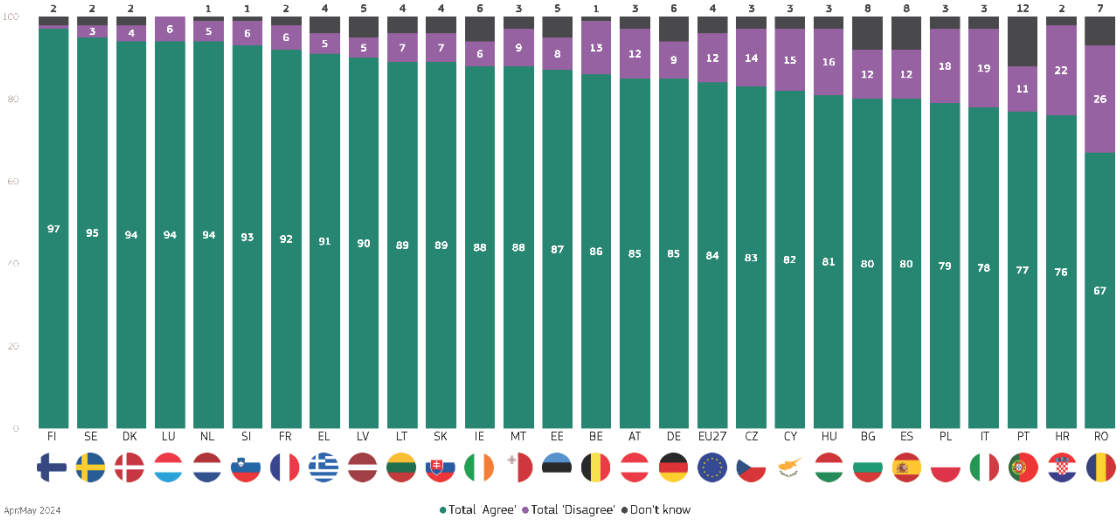
For each of these statements, less than one in ten respondents say they don't know (between 4% and 9%).

[22] QB6. Please tell me to what extent you agree or disagree with each of the following statements. 2) Robots and Artificial Intelligence are a good thing for society, because they help people do their jobs or carry out daily tasks at home; 3) Robots and Artificial Intelligence are technologies that require careful management; 9) Robots and Artificial Intelligence should be used more widely outside the workplace. Totally agree; Tend to agree; Tend to disagree; Totally disagree; Don't know.

In 22 EU Member States, at least eight in ten agree that robots and Artificial Intelligence are technologies that require careful management. Agreement levels are highest in Finland (97%) and Sweden (95%) as well as in Denmark, Luxembourg and the Netherlands (all 94%).

At the opposite end of the scale, around two thirds agree in Romania (67%), as do 76% in Croatia and 77% in Portugal.

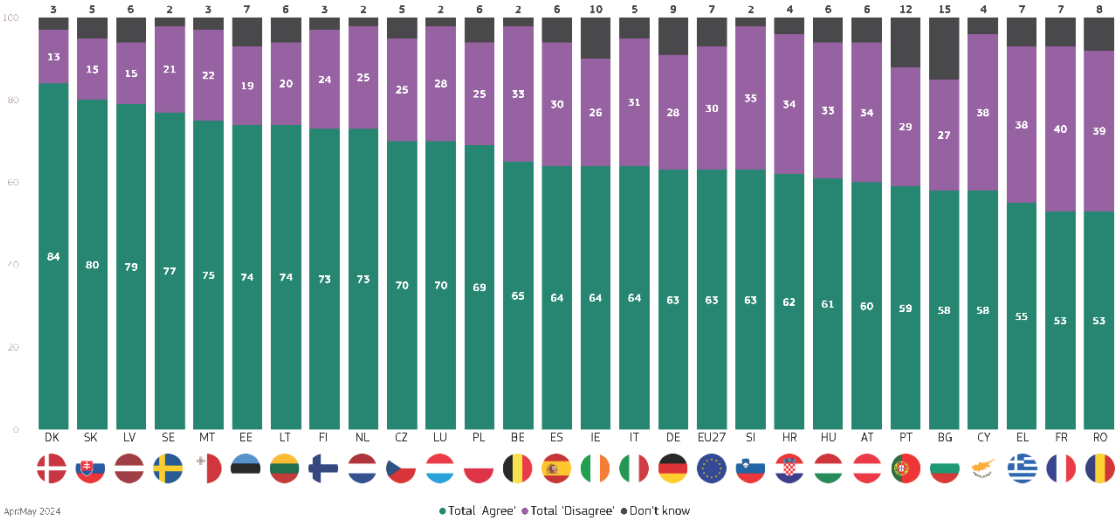
QB6.3. Please tell me to what extent you agree or disagree with each of the following statements.:-Robots and Artificial Intelligence are technologies that require careful management (%)



In 21 countries, at least six in ten believe that robots and Artificial Intelligence are a good thing for society, because they help people do their jobs or carry out daily tasks at home. This share of respondents is highest in Denmark (84%), Slovakia (80%) and Latvia (79%).

At the other end of the spectrum, 53% in France and Romania and 55% in Greece give this answer.

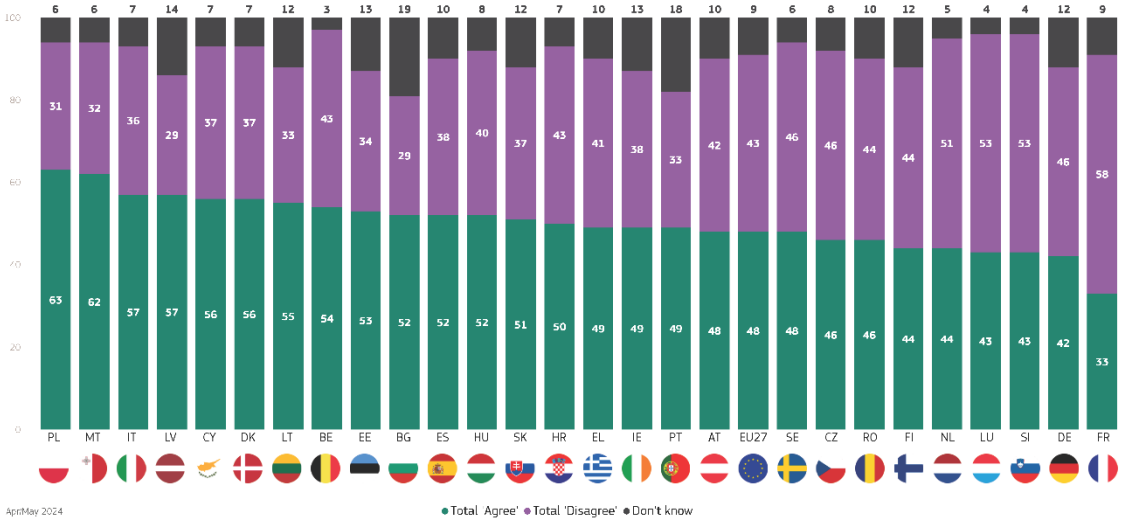
QB6.2. Please tell me to what extent you agree or disagree with each of the following statements.:-Robots and Artificial Intelligence are a good thing for society, because they help people do their jobs or carry out daily tasks at home (%)



A majority of respondents in 13 EU Member States agree that robots and Artificial Intelligence should be used more widely outside the workplace. This is especially the case in Poland (63%) and Malta (62%) and in Italy and Latvia (both 57%).

At the opposite end of the scale, one third hold this view in France (33%), as do 42% in Germany and

QB6.9. Please tell me to what extent you agree or disagree with each of the following statements.:-Robots and Artificial Intelligence should be used more widely outside the workplace (%)



b) On specific activities

Perceptions of the use of digital technologies, including Artificial Intelligence, for activities in the workplace are mixed: slightly more negative than **positive, with the exception of improving workers'** safety and security, allocating tasks and managing working schedules and shifts.

Around two thirds of the respondents (67%) positively perceive the use of digital technologies, including Artificial Intelligence, to **improve workers' safety and security**. This compares to 25% who have a negative perception[23].

Less than half (49%) – yet a relative majority – positively perceive the use of such technologies to allocate tasks to workers or manage their working schedules and shifts, while 44% negatively perceive this.

Minorities of the respondents positively perceive the use of such technologies for the other activities tested in the survey. Positive perceptions of the use of digital technologies to **collect, process, and store workers'** personal data are shared by 44% of the respondents, compared to 49% who hold a negative view.

Similarly, 43% positively perceive the use of these technologies to gather additional information on applicants for a job, while half of the respondents see this in a negative light.

The use of digital technologies to select applicants for a job is positively regarded by 36% of the respondents, compared to 57% who negatively perceive their use for this activity.

The same figures can be observed with regard to perceptions of the use of these technologies to assess **workers' performance**: 36% have positive perceptions and 57% have negative perceptions.



Around three in ten (31%) hold positive perceptions towards the use of such technologies to monitor workers. Conversely, more than six in ten (63%) perceive this in a negative manner.

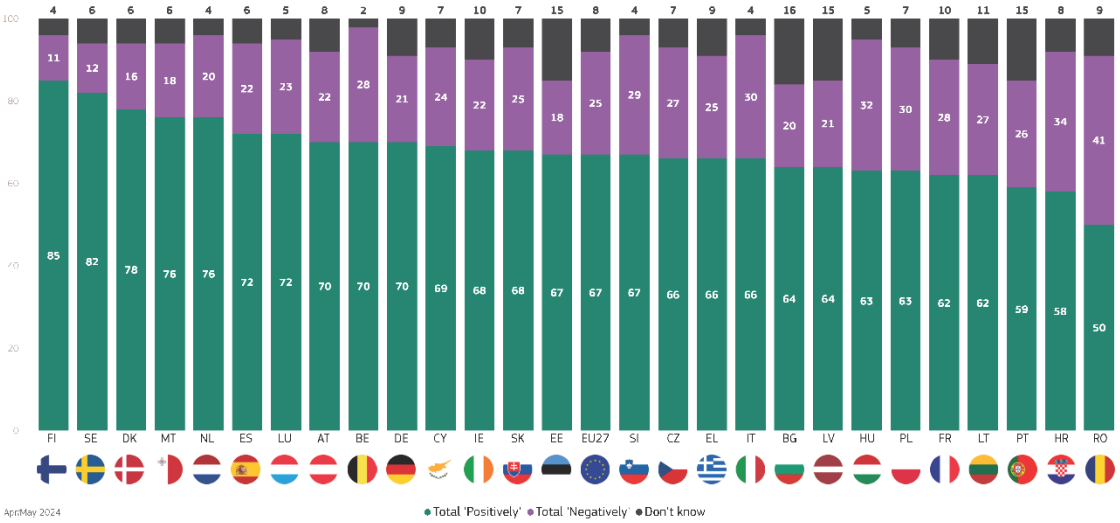
Lastly, less than one in five (16%) positively perceive the use of digital technologies to automatically fire workers, compared to 78% who regard this in a negative manner.

[23] QB8. In general, how positively or negatively do you perceive the use of digital technologies, including Artificial Intelligence, in the workplace for each of the following activities: 1) Gathering additional information on applicants for a job; 2) Selecting applicants for a job; 3) Allocating tasks to workers or managing their working schedules and shifts; 4) Collecting, processing, and storing workers' personal data; 5) Improving workers' safety and security; 6) Monitoring workers; 7) Assessing workers' performance; 8) Automatically firing workers. Very positively; Somewhat positively; Somewhat negatively; Very negative; Don't know.

In all EU Member States, at least half of the respondents positively perceive the use of digital technologies to **improve workers' safety and security**. This view is most widely held in Finland (85%), Sweden (82%) and Denmark (78%).

At the other end of the spectrum, half of the respondents positively regard this in Romania and close to six in ten do so in Croatia (58%) and Portugal (59%).

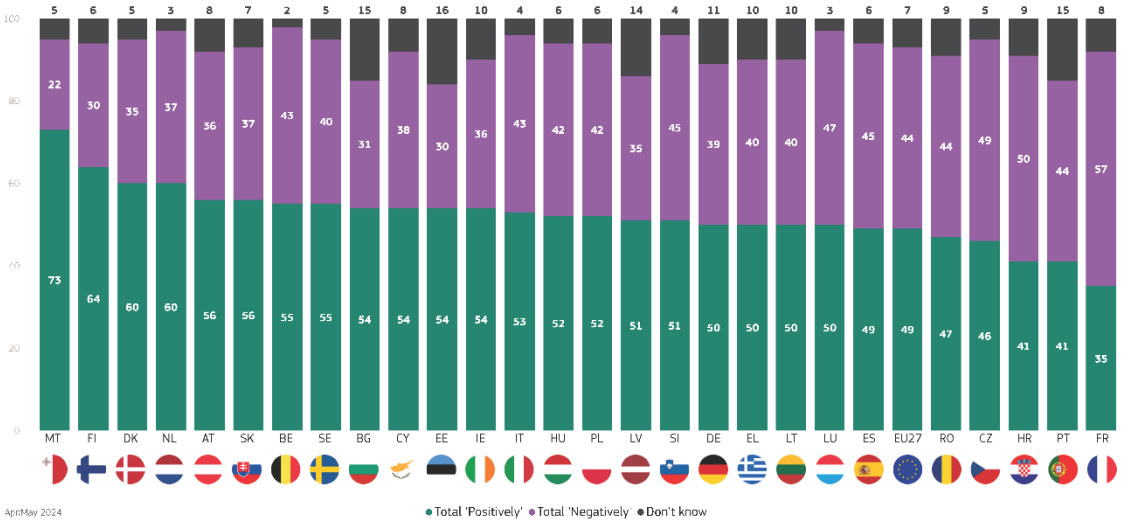
QB8.5. In general, how positively or negatively do you perceive the use of digital technologies, including Artificial Intelligence, in the workplace for each of the following activities:-Improving workers' safety and security (%)



Positive perceptions of the use of digital technologies to allocate tasks to workers or manage their working schedules and shifts are shared by at least half of the respondents in 21 countries.

This proportion ranges from at least six in ten in Malta (73%) and Finland (64%) and in Denmark and the Netherlands (both 60%) to 35% in France and 41% in Croatia and Portugal.

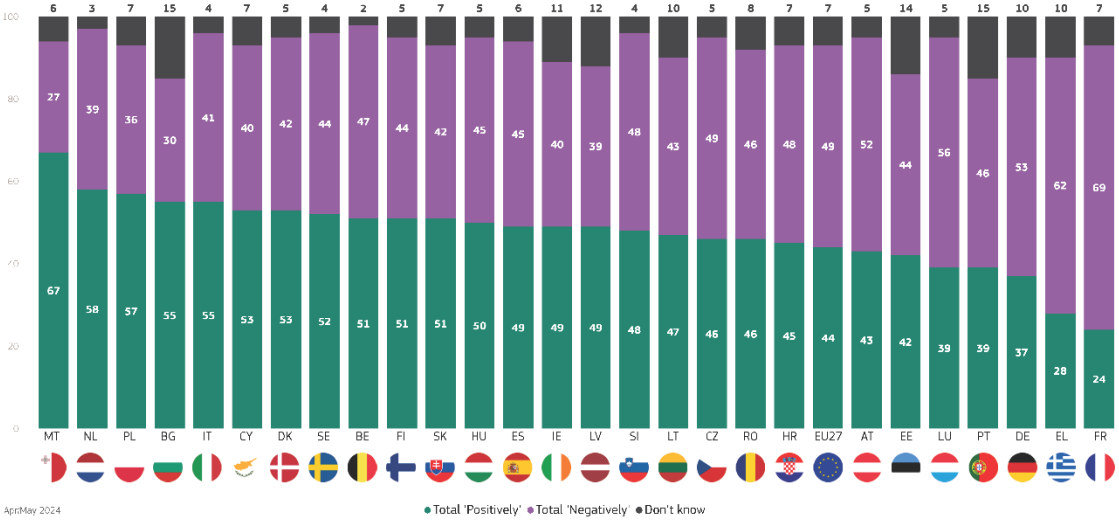
QB8.3. In general, how positively or negatively do you perceive the use of digital technologies, including Artificial Intelligence, in the workplace for each of the following activities:-Allocating tasks to workers or managing their working schedules and shifts (%)



In 12 EU Member States, at least half see the use of digital technologies to collect, process, and store workers' personal data in a positive light. This view is most widespread in Malta (67%), the Netherlands (58%) and Poland (57%).

At the opposite end of the scale, respondents in France (24%), Greece (28%) and Germany (37%) are the least likely to hold this opinion.

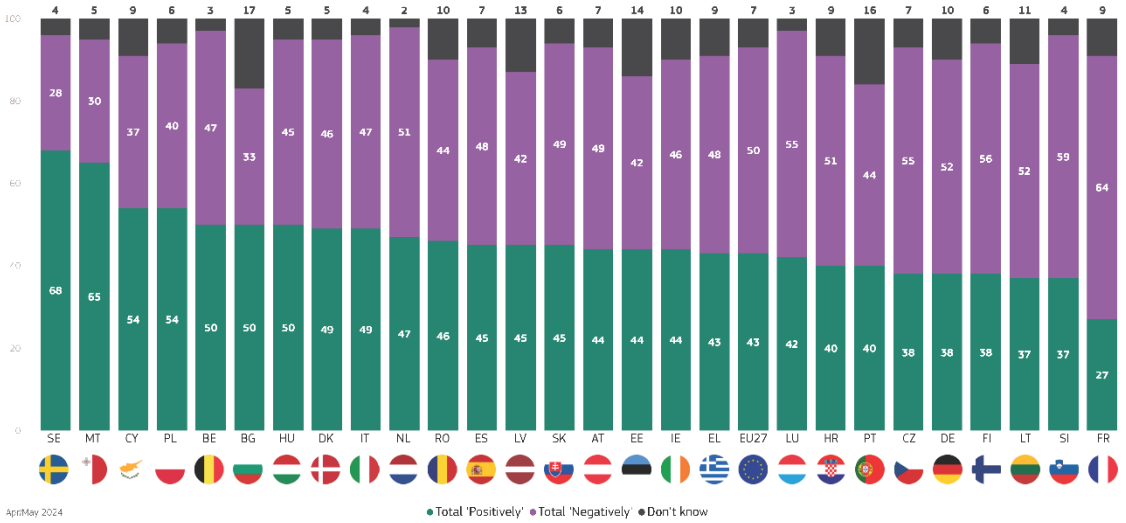
QB8.4. In general, how positively or negatively do you perceive the use of digital technologies, including Artificial Intelligence, in the workplace for each of the following activities:-Collecting, processing, and storing workers' personal data (%)



At least four in ten in 21 countries positively view the use of digital technologies to gather additional information on applicants for a job. A majority of respondents give this answer in Sweden (68%) and Malta (65%) and in Cyprus and Poland (both 54%).

This compares to 27% in France and 37% in Lithuania and Slovenia who hold a positive view.

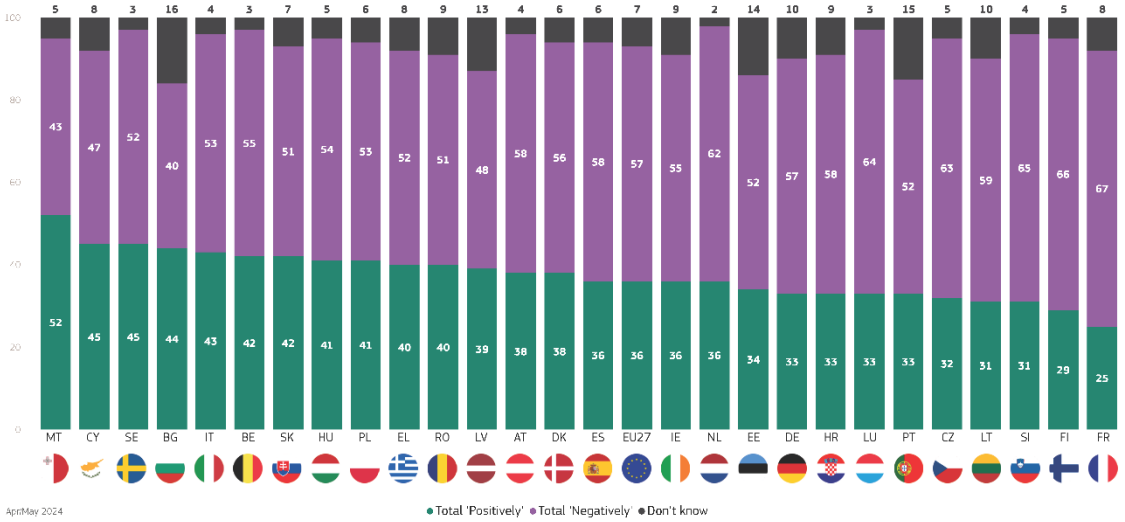
QB8.1. In general, how positively or negatively do you perceive the use of digital technologies, including Artificial Intelligence, in the workplace for each of the following activities:-Gathering additional information on applicants for a job (%)



In 11 EU Member States, the use of digital technologies to select applicants for a job is positively perceived by at least four in ten. This is most notably the case for a majority in Malta (52%) and for 45% in Cyprus and Sweden.

While the lowest proportions holding this view can be observed in France (25%) and Finland (29%) and in Lithuania and Slovenia (both 31%).

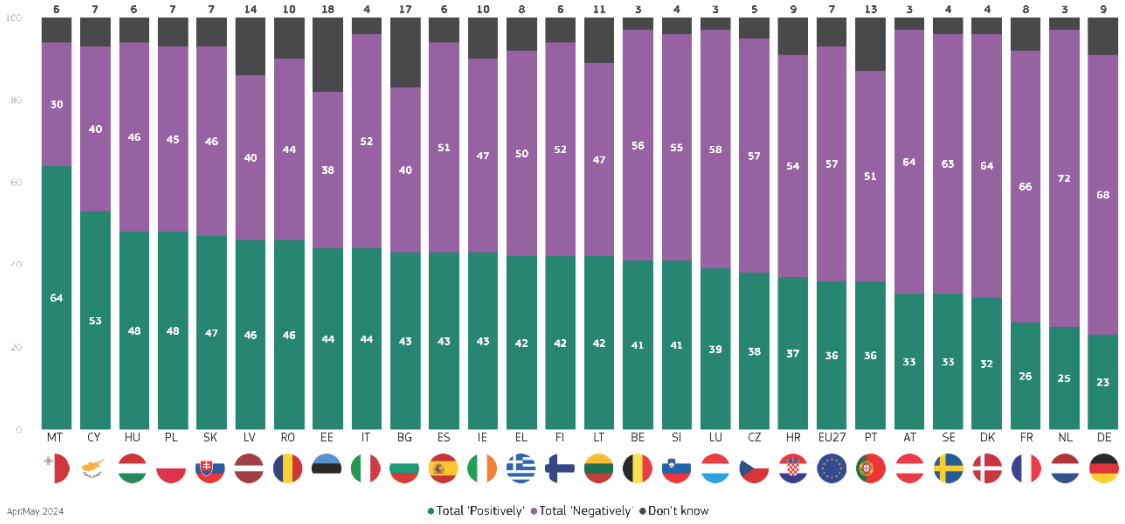
QB8.2. In general, how positively or negatively do you perceive the use of digital technologies, including Artificial Intelligence, in the workplace for each of the following activities:-Selecting applicants for a job (%)



In 17 countries, more than four in ten positively perceive the use of digital technologies to **assess workers'** performance. This is the majority view in Malta (64%) and Cyprus (53%), while close to half say this in Hungary and Poland (both 48%).

At the other end of the scale, 23% in Germany, 25% in the Netherlands and 26% in France give this answer.

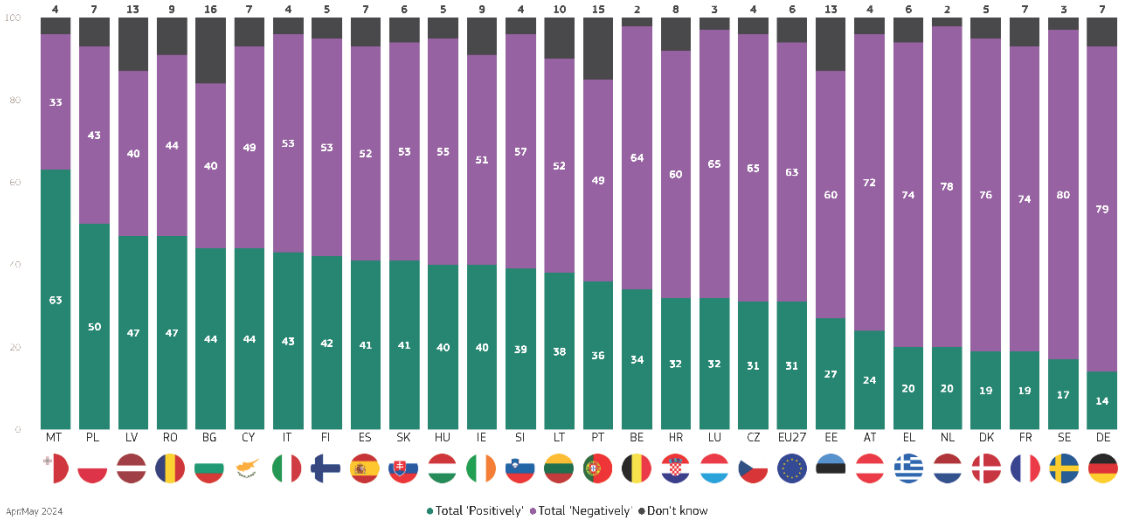
QB8.7. In general, how positively or negatively do you perceive the use of digital technologies, including Artificial Intelligence, in the workplace for each of the following activities:-Assessing workers' performance (%)



At least four in ten in 12 countries regard the use of digital technologies to monitor workers in a positive light.

This proportion ranges from 63% in Malta, 50% in Poland and 47% in Latvia and Romania to less than one fifth in Germany (14%) and Sweden (17%) and in Denmark and France (both 19%).

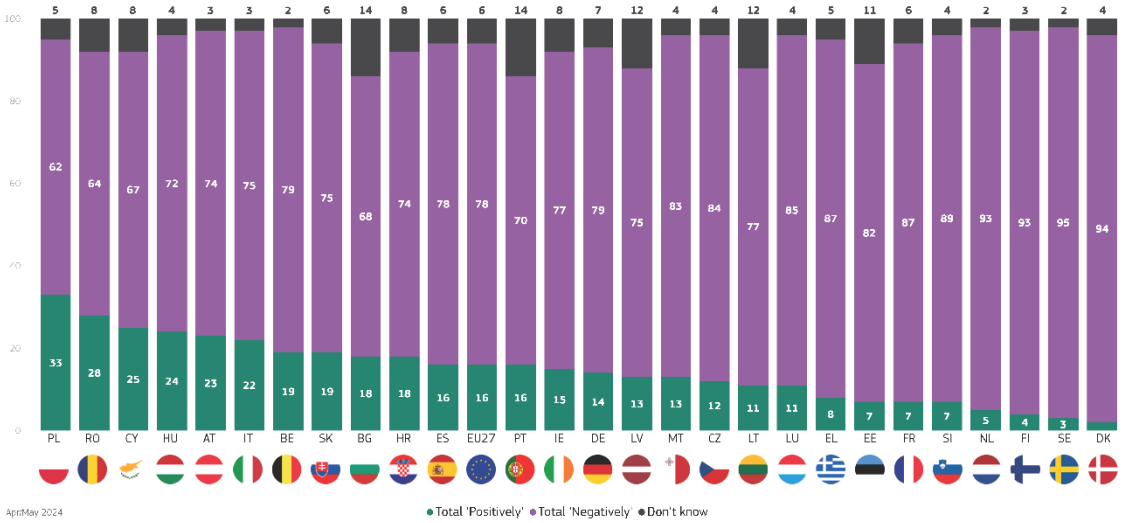
QB8.6. In general, how positively or negatively do you perceive the use of digital technologies, including Artificial Intelligence, in the workplace for each of the following activities:-Monitoring workers (%)



Interestingly, more than two in ten in six EU Member States positively regard the use of digital technologies to automatically fire workers, including at least one quarter in Poland (33%), Romania (28%) and Cyprus (25%).

This compares to less than one in twenty who hold this view in Denmark (2%), Sweden (3%) and Finland (4%).

QB8.8. In general, how positively or negatively do you perceive the use of digital technologies, including Artificial Intelligence, in the workplace for each of the following activities:-Automatically firing workers (%)



3. Perceptions of rules on digital technologies in the workplace

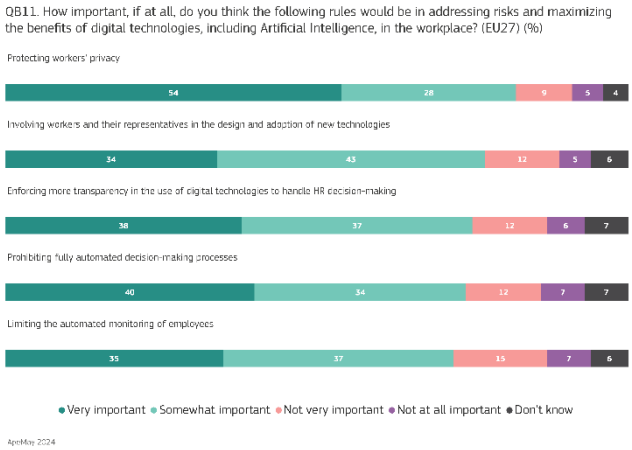
Large majorities attach importance to rules addressing risks and maximising the benefits of digital technologies in the workplace[24]

More than eight in ten (82%) say protecting workers' privacy would be important, with more than half (54%) who think this would be 'very important'. Conversely, 14% believe this rule would not be important.

Involving workers and their representatives in the design and adoption of new technologies is considered by more than three quarters (77%) as an important measure, with 34% who think this would be 'very important'. Less than one in five (17%) do not attach importance to this measure.

Three quarters of the respondents think that enforcing more transparency in the use of digital technologies to handle HR decision-making would be important, including 38% who consider this as 'very important', while close to one in five (18%) say this would not be important.

Prohibiting fully automated decision-making processes is considered important to address risks and maximising the benefits of digital technologies in the workplace by close to three quarters (74%), including 40% who regard this as 'very important'. By contrast, nearly one in five (19%) say this would not be important.



More than seven in ten (72%) think it would be important to limit the automated monitoring of employees, including 35% who see this as 'very important', while around one in five (22%) do not attach importance to this measure.

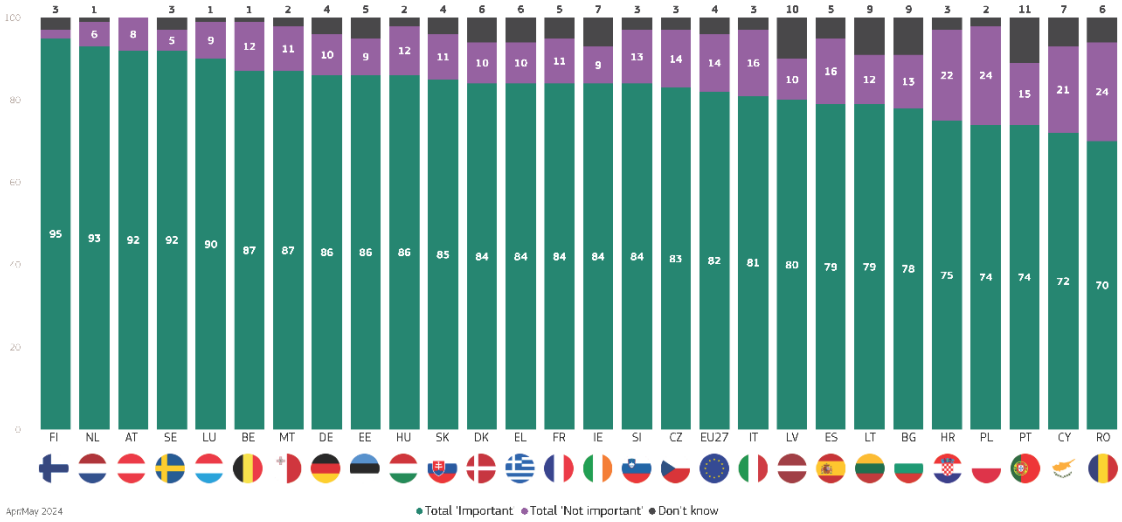
For all these measures, between 4% and 7% of the respondents say they don't know whether they would be important.

[24] QB11. How important, if at all, do you think the following rules would be in addressing risks and maximizing the benefits of digital technologies, including Artificial Intelligence, in the workplace? 1) Protecting workers' privacy; 2) Prohibiting fully automated decision-making processes; 3) Limiting the automated monitoring of employees; 4) Enforcing more transparency in the use of digital technologies to handle HR decision-making; 5) Involving workers and their representatives in the design and adoption of new technologies. Very important; Somewhat important; Not very important; Not at all important; Don't know.

In 19 EU Member States, at least eight in ten think protecting workers' privacy would be important in addressing risks and maximising the benefits of digital technologies in the workplace. More than nine in ten think this in Finland (95%) and the Netherlands (93%) and in Austria and Sweden (both 92%).

At the other end of the spectrum, less than three quarters hold this view in Romania (70%) and Cyprus (72%) and in Poland and Portugal (both 74%).

QB11.1. How important, if at all, do you think the following rules would be in addressing risks and maximizing the benefits of digital technologies, including Artificial Intelligence, in the workplace?:-Protecting workers' privacy (%)

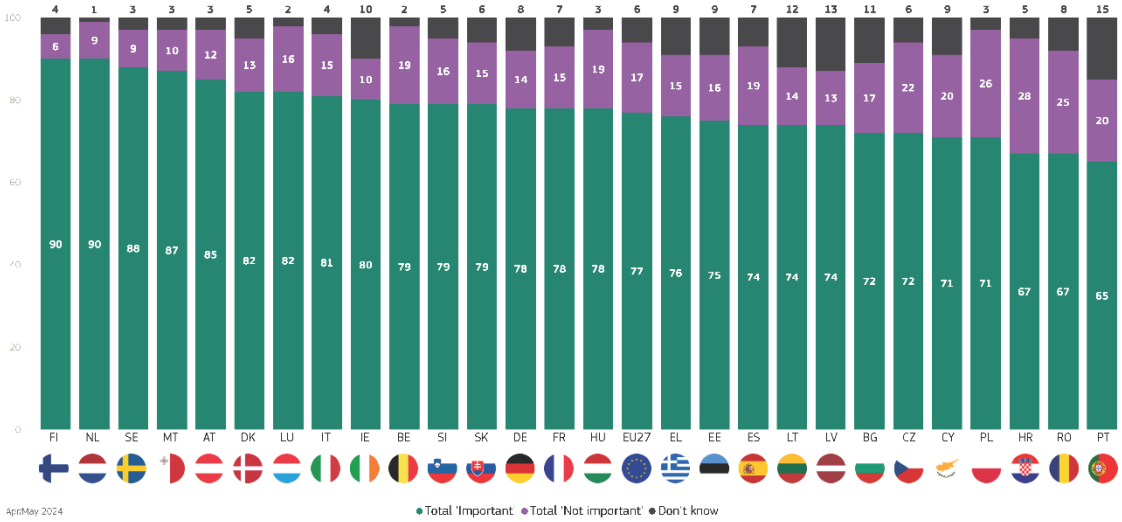


At least three quarters in 17 countries indicate that involving workers and their representatives in the design and adoption of new technologies would be important in addressing risks and maximizing the benefits of digital technologies in the workplace.

This proportion ranges from 90% in Finland and the Netherlands and 88% in Sweden to 65% in Portugal and 67% in Croatia and Romania.

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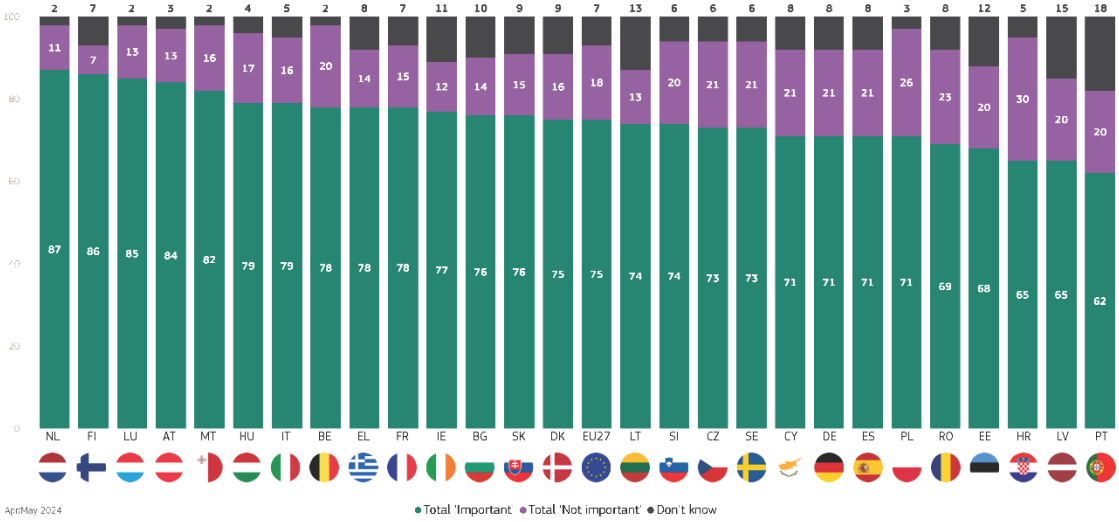
QB11.5. How important, if at all, do you think the following rules would be in addressing risks and maximizing the benefits of digital technologies, including Artificial Intelligence, in the workplace?:-Involving workers and their representatives in the design and adoption of new technologies (%)



In 14 EU Member States, at least three quarters attach importance to enforcing more transparency in the use of digital technologies to handle HR decision-making in addressing risks and maximizing the benefits of digital technologies in the workplace.

The largest shares of respondents who give this answer are found in the Netherlands (87%), Finland (86%) and Luxembourg (85%), while the lowest can be observed in Portugal (62%) and in Croatia and Latvia (both 65%).

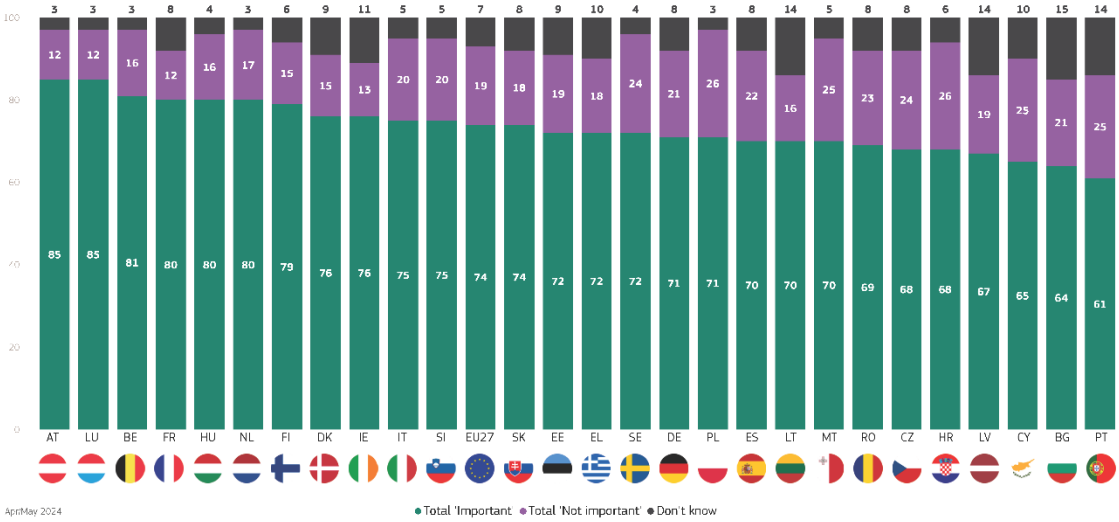
QB11.4. How important, if at all, do you think the following rules would be in addressing risks and maximizing the benefits of digital technologies, including Artificial Intelligence, in the workplace?:-Enforcing more transparency in the use of digital technologies to handle HR decision-making (%)



In 20 countries, at least seven in ten consider that prohibiting fully automated decision-making processes would be important in addressing risks and maximising the benefits of digital technologies in the workplace.

This view is most widely held in Austria and Luxembourg (both 85%) and in Belgium (81%), while respondents in Portugal (61%), Bulgaria (64%) and Cyprus (65%) are the least likely to share this opinion.

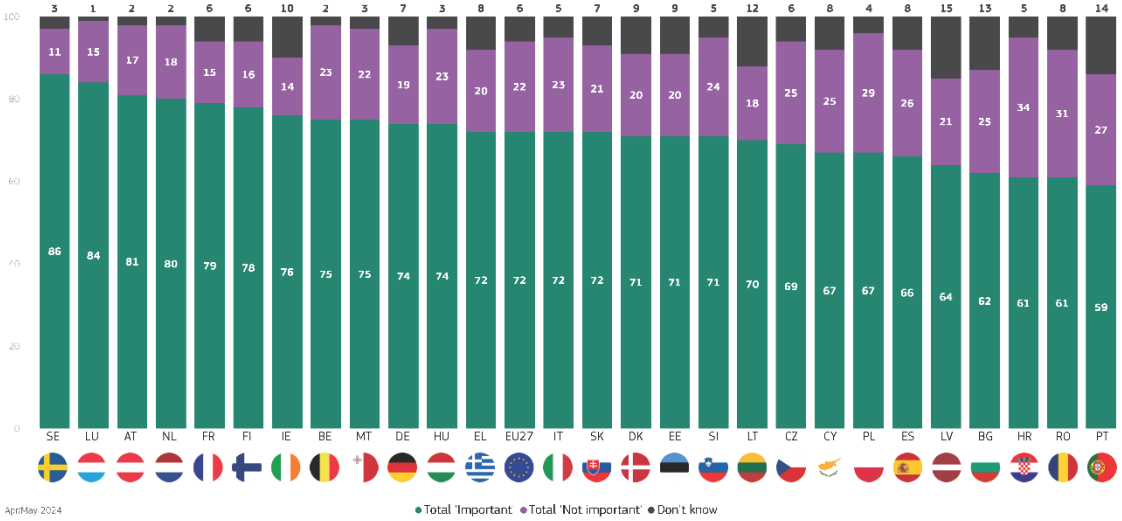
QB11.2. How important, if at all, do you think the following rules would be in addressing risks and maximizing the benefits of digital technologies, including Artificial Intelligence, in the workplace?:-Prohibiting fully automated decision-making processes (%)



At least seven in ten in 18 countries believe limiting the automated monitoring of employees would be important to address risks and maximising the benefits of digital technologies in the workplace.

More than eight in ten give this answer in Sweden (86%), Luxembourg (84%) and Austria (81%), while 59% do so in Portugal and 61% in Croatia and Romania.

QB11.3. How important, if at all, do you think the following rules would be in addressing risks and maximizing the benefits of digital technologies, including Artificial Intelligence, in the workplace?–Limiting the automated monitoring of employees (%)





IV. Experience with digital technologies in the workplace

1. Awareness of the use of digital technologies

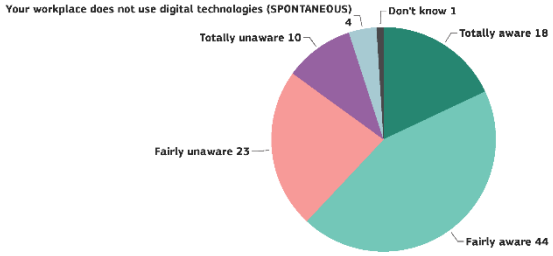
a) Overall

More than six in ten employees are aware of the use that their employer makes of digital technologies, to manage their activities[25]

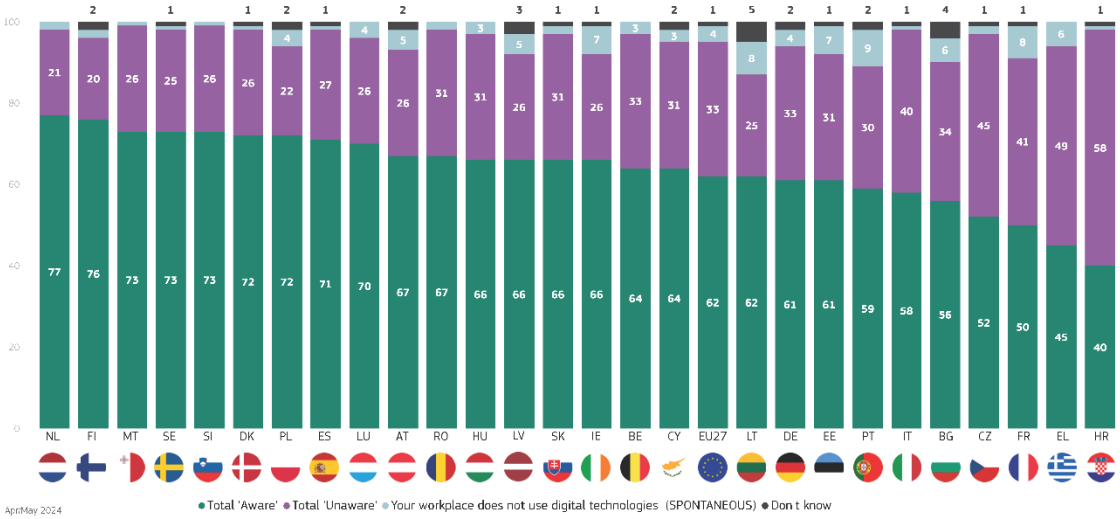
More than six in ten respondents who are currently in employment (62%) say they are aware of the use of these technologies, with 18% who are 'totally aware'. Conversely, more than three in ten (33%) indicate that they are unaware, including one in ten who are 'totally unaware'. Less than one in twenty (4%) spontaneously say that their workplace does not use digital technologies and 1% say they don't know.

In 24 out of the 27 EU Member States, a majority of respondents currently in employment say they are aware of the use that their employer makes of digital technologies, including Artificial Intelligence, to manage their or their co-workers' activities. This proportion is the highest in the Netherlands (77%) and Finland (76%) and in Malta, Slovenia and Sweden (all 73%). At the opposite end of the scale, half of these respondents or less are aware in Croatia (40%), Greece (45%) and France (50%).

QB4. Overall, how aware are you of the use that your employer makes of digital technologies, including artificial intelligence, to manage your or your co-workers' activities? Examples include setting working schedules, monitoring your activities, or evaluating performance. (EU27) (%)



QB4. Overall, how aware are you of the use that your employer makes of digital technologies, including artificial intelligence, to manage your or your co-workers' activities? Examples include setting working schedules, monitoring your activities, or evaluating performance. (%)



[25] QB4. Overall, how aware are you of the use that your employer makes of digital technologies, including artificial intelligence, to manage your or your co-workers' activities? Examples include setting working schedules, monitoring your activities, or evaluating performance. Totally aware; Fairly aware; Fairly unaware; Totally unaware; Your workplace does not use digital technologies (SPONTANEOUS); Don't know.

b) Activities performed by digital technologies

More than one in five respondents with a current or past occupation say that most of the activities tested in the survey have been performed by digital technologies in their workplace

Respondents who have a current occupation or who have had one in the past were asked whether a series of activities have ever been performed by digital technologies, including Artificial Intelligence, in their current or previous workplaces[26].

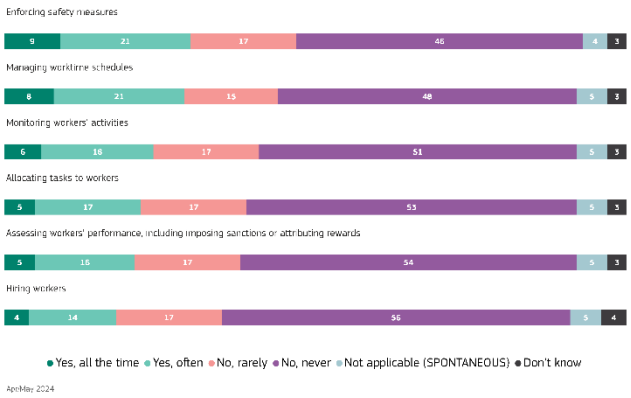
Three in ten of these respondents indicate that enforcing safety measures has been performed by digital technologies in their workplaces, with 9% saying this has been done 'all the time' and 21% 'often'. Conversely, more than six in ten (63%) say that this activity has not been performed by digital technologies, with 46% stating that this has 'never' occurred and 17% that this has 'rarely' occurred.

A similar proportion (29%) reports that managing worktime schedules has been performed by these technologies in their workplaces. Within this proportion, 8% saying this has happened 'all the time' and 21% 'often'. Again, 63% indicate that this activity has not been performed by digital technologies, with 48% who say this has 'never' been done and 15% who say this has 'rarely' been done.

Monitoring workers' activities is said to have been performed by digital technologies by around one quarter (24%), with 6% saying this has occurred 'all the time' and 18% 'often'. By contrast, close to seven in ten (68%) report that this activity has not been performed by such technologies (51% 'never' and 17% 'rarely').

More than two in ten (22%) say that digital technologies have allocated tasks to workers in their workplaces, including one in twenty who indicate this has happened 'all the time' and 17% who say this has happened 'often'. Seven in ten indicate that this activity has not been performed by these technologies (53% 'never' and 17% 'rarely').

QB7. Have the following activities ever been performed by digital technologies, including Artificial Intelligence, in your current or previous workplaces? For each activity, please refer to your most recent workplace experience. (EU27) (%)



Around one in five (21%) state assessing workers' performance, including imposing sanctions or attributing rewards, has been performed by digital technologies. Within this proportion, 5% say this has been done 'all the time' and 16% 'often'. Around seven in ten (71%) report that digital technologies have not performed this task (54% 'never' and 17% 'rarely').

Finally, 18% indicate that digital technologies have been used to hire workers, with 4% reporting this has occurred 'all the time' and 14% 'often'. However, close to three quarters (73%) say this activity has not been performed by digital technologies in their workplaces (56% 'never' and 17% 'rarely').

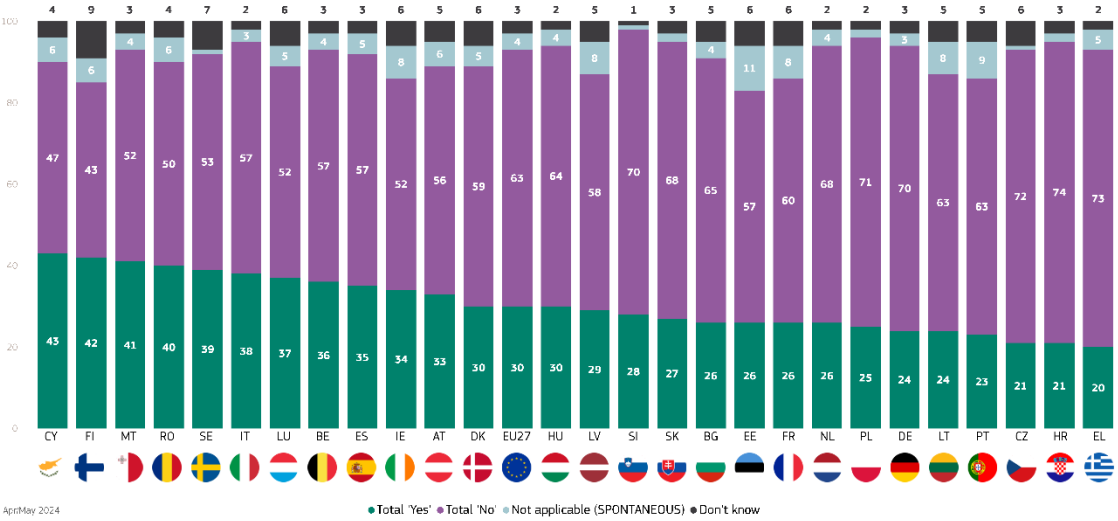
For all these activities, between 3% and 4% say they don't know whether they have been performed by digital technologies.

[26] QB7. Have the following activities ever been performed by digital technologies, including Artificial Intelligence, in your current or previous workplaces? For each activity, please refer to your most recent workplace experience. 1) Hiring workers; 2) Allocating tasks to workers; 3) Managing worktime schedules; 4) Monitoring workers' activities; 5) Assessing workers' performance, including imposing sanctions or attributing rewards; 6) Enforcing safety measures. Yes, all the time; Yes, often; No, rarely; No, never; Not applicable (SPONTANEOUS); Don't know.

In 13 EU Member States, at least three in ten respondents who have a current or past occupation say that that enforcing safety measures has been performed by digital technologies, including Artificial Intelligence, in their current or previous workplaces.

This proportion ranges from more than four in ten in Cyprus (43%), Finland (42%) and Malta (41%) to 20% in Greece and 21% in Croatia and Czechia.

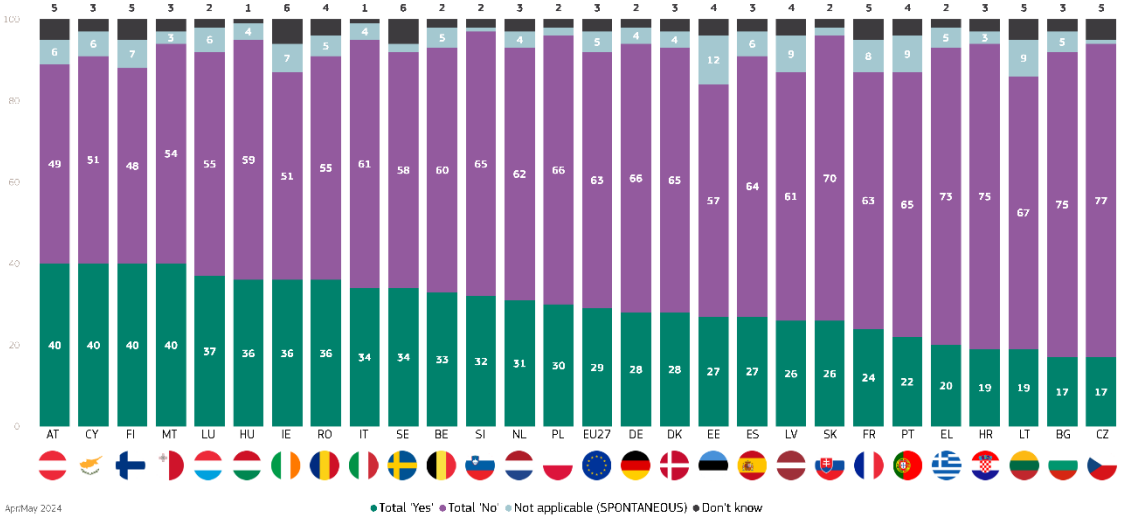
QB7.6. Have the following activities ever been performed by digital technologies, including Artificial Intelligence, in your current or previous workplaces? For each activity, please refer to your most recent workplace experience.:-Enforcing safety measures (%)



At least three in ten in 14 countries report that digital technologies, including Artificial Intelligence, have performed the management of worktime schedules in their workplaces, with four in ten saying this in Austria, Cyprus, Malta and Finland.

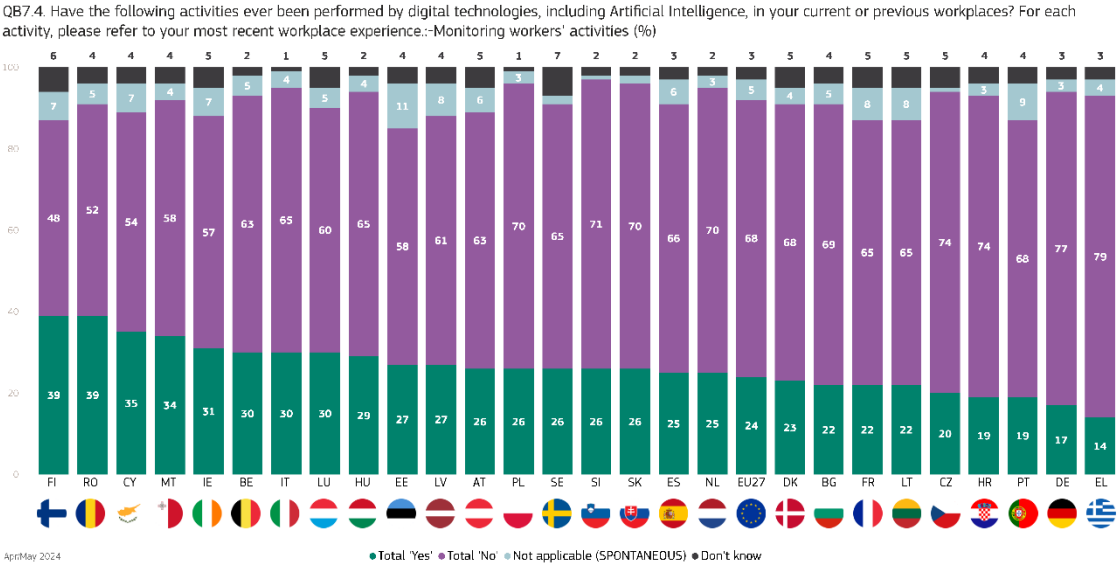
At the opposite end of the scale, less than one in five give this answer in Bulgaria and Czechia (both 17%) and in Croatia and Lithuania (both 19%).

QB7.3. Have the following activities ever been performed by digital technologies, including Artificial Intelligence, in your current or previous workplaces? For each activity, please refer to your most recent workplace experience.:-Managing worktime schedules (%)



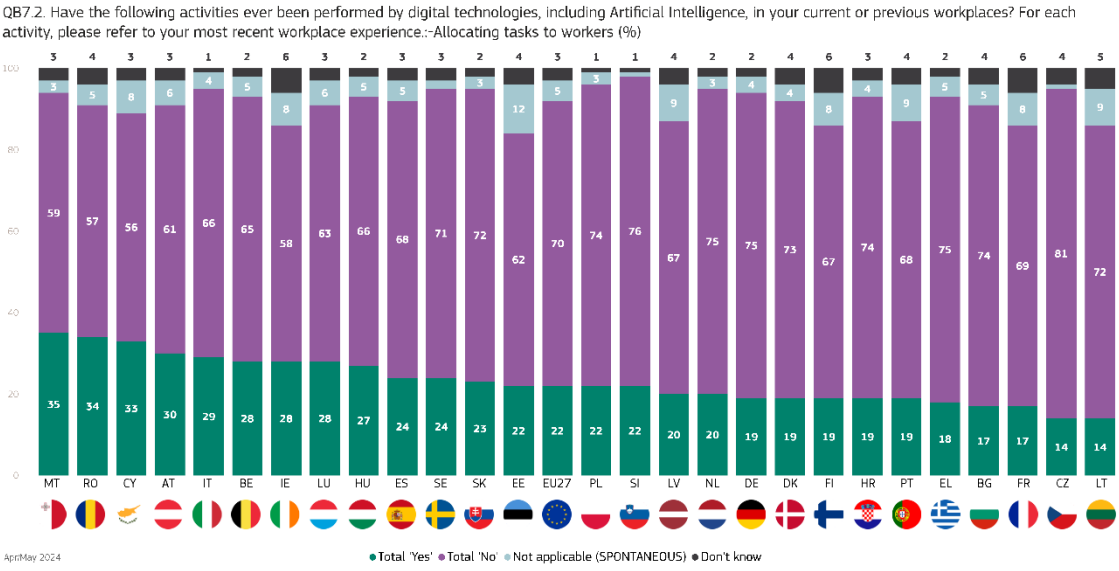
Monitoring workers' activities is said to have been performed by digital technologies, including Artificial Intelligence, by more than one quarter in 16 countries. The largest shares giving this answer are found in Romania and Finland (both 39%) as well as in Cyprus (35%).

This compares to less than two in ten who say this in Greece (14%) and Germany (17%) and in Croatia and Portugal (both 19%).



In nine EU Member States, more than one quarter of respondents currently or previously employed say that digital technologies, including Artificial Intelligence, have allocated tasks to workers in their current or previous workplaces.

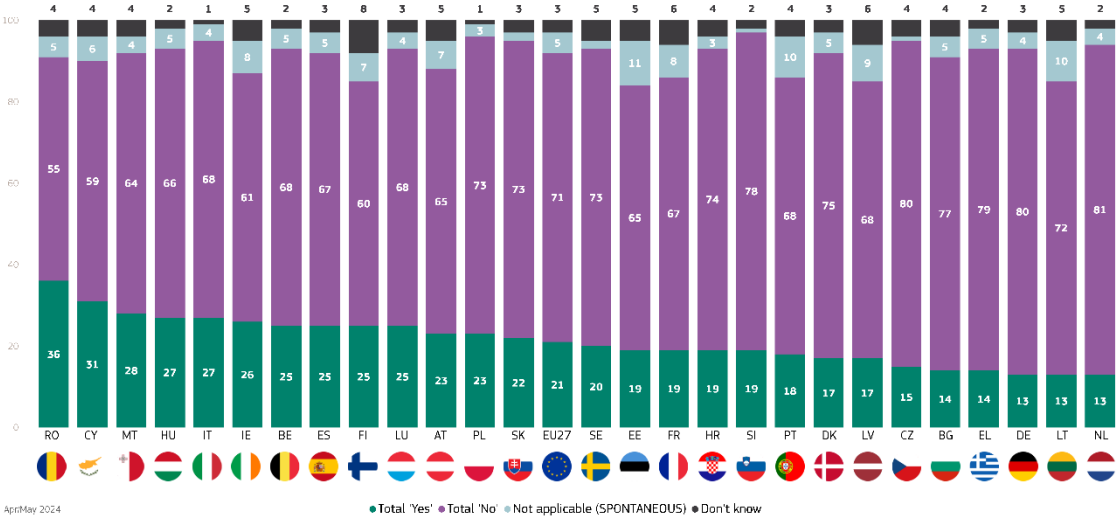
More than three in ten give this answer in Malta (35%), Romania (34%) and Cyprus (33%). At the other end of the spectrum, 14% in Czechia and Lithuania and 17% in Bulgaria and France report this.



At least one quarter in ten EU Member States indicate that digital technologies, including Artificial Intelligence, have assessed workers' performance, including imposing sanctions or attributing rewards, in their workplaces.

Those in Romania (36%), Cyprus (31%) and Malta (28%) are the most likely to report this, while those in Germany, the Netherlands and Lithuania (all 13%) are the least likely to do so.

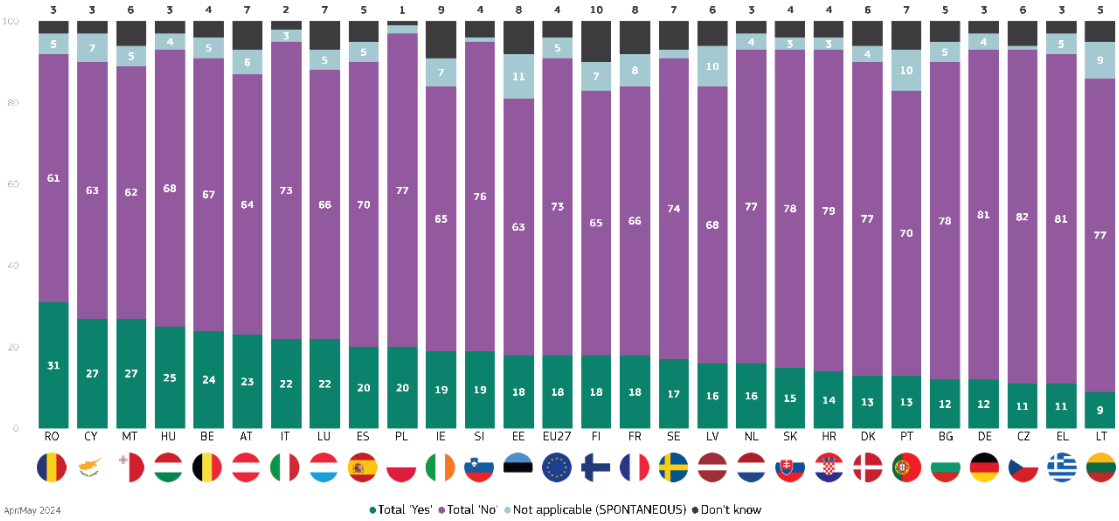
QB7.5. Have the following activities ever been performed by digital technologies, including Artificial Intelligence, in your current or previous workplaces? For each activity, please refer to your most recent workplace experience.:-Assessing workers' performance, including imposing sanctions or attributing rewards (%)



In ten countries, at least one fifth say that hiring workers has been performed by digital technologies, including Artificial Intelligence, in their workplaces, most notably in Romania (31%) and in Cyprus and Malta (both 27%).

Conversely, 9% in Lithuania indicate this, as do 11% in Czechia and Greece.

QB7.1. Have the following activities ever been performed by digital technologies, including Artificial Intelligence, in your current or previous workplaces? For each activity, please refer to your most recent workplace experience.:-Hiring workers (%)



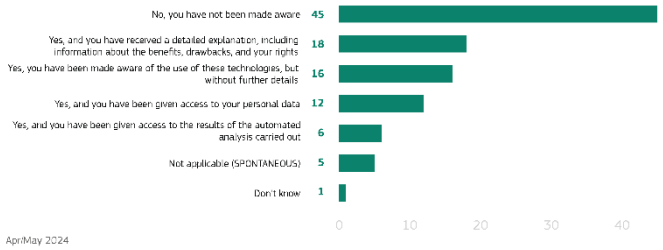
c) Communication around the use of digital technologies

Close to half of those currently in employment say their employer informed them about the use of digital technologies[27].

Among these respondents, 16% say they have been made aware of the use of these technologies, but without further details. Moreover, 18% report having received a detailed explanation, including information about the benefits, drawbacks, and their rights, and 12% indicate they have been given access to their personal data. In addition, 6% have been given access to the results of the automated analysis carried out. Overall, close to half (49%) say their employer informed them about the use of digital technologies to manage activities in their workplace.

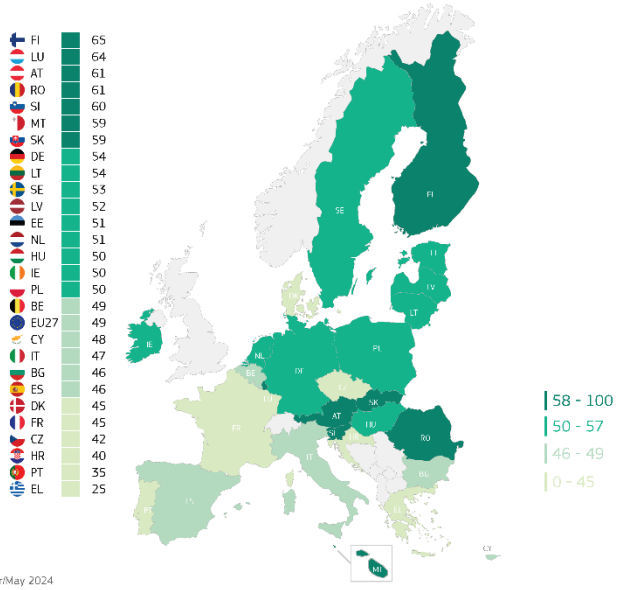
By contrast, 45% say they have not been made aware, while one in twenty spontaneously say the question is not applicable and 1% say they don't know.

QB9. Did your employer inform you about the use of digital technologies, including Artificial Intelligence, to manage activities in your workplace? (EU27) (%)



Apr/May 2024

QB9. Did your employer inform you about the use of digital technologies, including Artificial Intelligence, to manage activities in your workplace? - Total 'Yes' (%)



Apr/May 2024

The country-level analysis shows that at least half of the employees in 16 EU Member States report having been informed by their employer about the use of digital technologies, including Artificial Intelligence, to manage activities in their workplace.

This share ranges from more than six in ten in Finland (65%) and Luxembourg (64%) and in Austria and Romania (both 61%) to four in ten or less in Greece (25%), Portugal (35%) and Croatia (40%).

In four countries, the majority says they have not been made aware by their employer: Greece (70%), Croatia (57%), Czechia (54%) and Spain (52%).

[27] QB9. Did your employer inform you about the use of digital technologies, including Artificial Intelligence, to manage activities in your workplace? Yes, you have been made aware of the use of these technologies, but without further details; Yes, and you have received a detailed explanation, including information about the benefits, drawbacks, and your rights; Yes, and you have been given access to your personal data; Yes, and you have been given access to the results of the automated analysis carried out; No, you have not been made aware; Not applicable (SPONTANEOUS); Don't know.

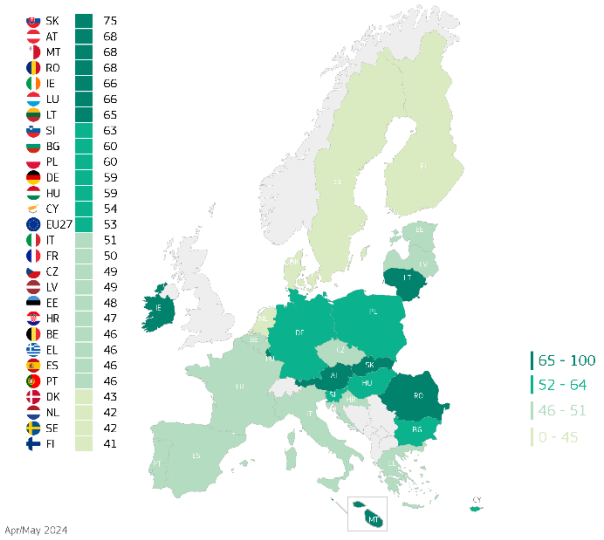
A majority of employers and managers say they have ensured that their employees or the people they manage have been informed about the use of digital technologies[28].

Mirroring the question asked to respondents currently in employment, those who are employers or managers were asked if they have ensured that their employees or the people they manage have been informed about the use of digital technologies, including Artificial Intelligence, to manage activities in their workplace

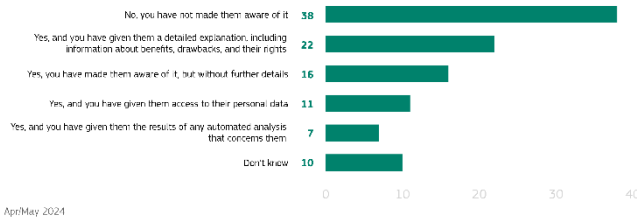
Among these respondents, 16% say they have made them aware of it, but without further details. More than two in ten (22%) indicate that they have given them a detailed explanation, including information about benefits, drawbacks, and their rights, while around one in ten (11%) report having given them access to their personal data. Finally, 7% state that they have given them the results of any automated analysis that concerns them. In total, 53% of employers and managers say they have ensured that their employees or the people they manage have been informed.

By contrast, less than four in ten (38%) report not having made them aware of the use of digital technologies to manage activities in their workplace, while 10% say they don't know.

QB10. Have you ensured that your employees or the people you manage have been informed about the use of digital technologies, including Artificial Intelligence, to manage activities in your workplace? - Total 'Yes' (%)



QB10. Have you ensured that your employees or the people you manage have been informed about the use of digital technologies, including Artificial Intelligence, to manage activities in your workplace? (EU27) (%)



Comparing the results of this question with the answers given by the employees, it can be observed that the proportion of employers and managers who self-report that information about the use of digital technologies was shared is slightly higher than the proportion of employees who report having been informed about this (53% vs 49%).

In particular, there is a four-percentage point gap between the share of employers and managers who say they have provided employees and people they manage with a detailed explanation, including information about the benefits, drawbacks, and their rights (22%), and the share of employees who report having received such explanation (18%). In addition, employers and managers are more likely than employees to say they don't know how to answer this question (10% vs 1%).

At country level, at least half of employers and managers in 15 EU Member States report having ensured that their employees or the people they manage have been informed about the use of digital technologies, including Artificial Intelligence, to manage activities in their workplace[29]. The highest proportions giving this answer are recorded in Slovakia (75%) and in Austria, Malta and Romania (all 68%), while the lowest are found in Finland (41%) and in the Netherlands and Sweden (both 42%).

Employers and managers in Greece (49%) and Croatia (48%) and in Denmark and Spain (both 47%) are the most likely to report that they have not made aware their employees or the people they manage of this information.

Close to one quarter or more in Finland (35%), Latvia (25%) and Sweden (24%) say they don't know.

[28] QB10. Have you ensured that your employees or the people you manage have been informed about the use of digital technologies, including Artificial Intelligence, to manage activities in your workplace? Yes, you have made them aware of it, but without further details; Yes, and you have given them a detailed explanation, including information about benefits, drawbacks, and their rights; Yes, and you have given them access to their personal data; Yes, and you have given them the results of any automated analysis that concerns them; No, you have not made them aware of it; Don't know.

[29] These results should be interpreted with caution, due to low sample sizes in some countries. In particular, sample sizes are less than 100 in Belgium, Cyprus, Greece, Luxembourg, Malta, Poland, Romania and Spain, while they are between 101 and 150 in Bulgaria, Croatia, Czechia, France, Hungary, Ireland, Italy, Latvia, Portugal, Slovakia and Slovenia.

2. Provision of tools and training to work effectively with digital technologies

A large majority of those currently in employment say their employer provides them with the necessary tools or training to work effectively with the most recent digital technologies

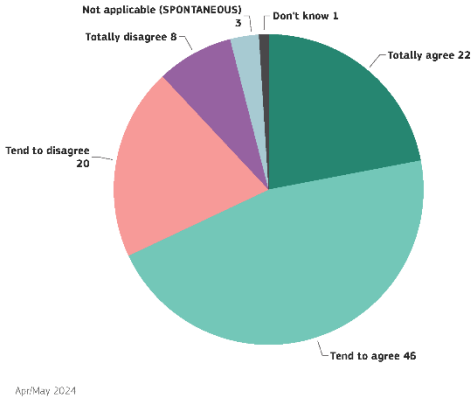
More than two thirds of the respondents currently in employment (68%) agree that their employer provides them with the necessary tools or training to work effectively with the most recent digital technologies, including Artificial Intelligence, with more than one in five (22%) who 'totally agree' with this statement[30].

Conversely, close to three in ten (28%) disagree that they are provided with the necessary tools or training, while 3% spontaneously say this question is 'not applicable' and 1% say they don't know.

A majority of employees in all 27 EU Member States report that their employer provides them with the necessary tools or training to work effectively with the most recent digital technologies, including Artificial Intelligence. The highest shares indicating this can be observed in Malta (80%) and the Netherlands (79%) and in Slovakia, Slovenia and Spain (all 78%). This compares to less than six in ten who agree in Greece (51%) and in Bulgaria and Romania (both 58%).

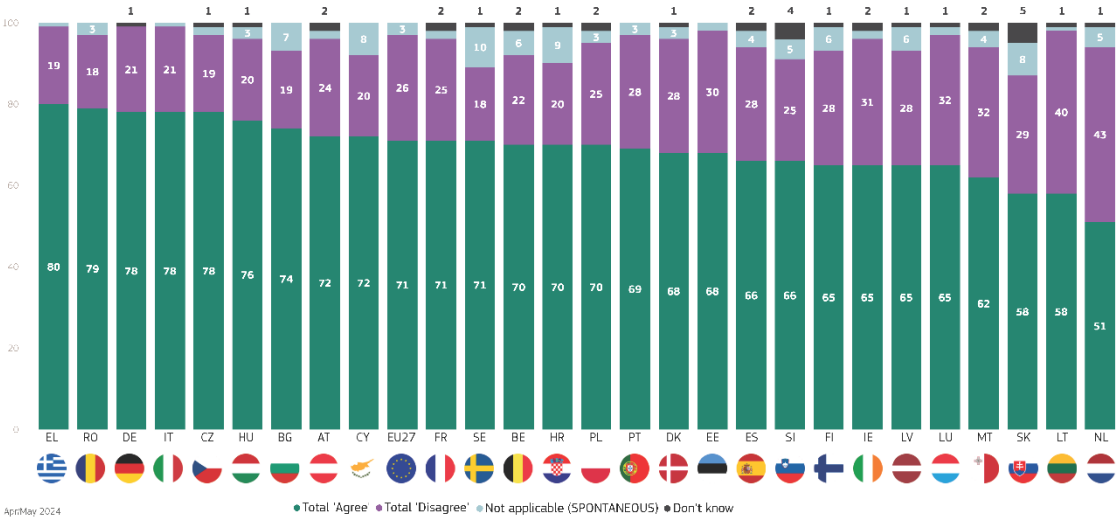
Those in Malta (55%) and the Netherlands (43%) and in Finland and Sweden (both 35%) are the most likely to 'totally agree' with this statement.

QB3. To what extent do you agree or disagree that your employer provides you with the necessary tools or training to work effectively with the most recent digital technologies, including Artificial Intelligence? (EU27) (%)



By contrast, the level of disagreement is highest in Greece (43%) and Romania (40%) as well as in Italy and Germany (both 32%).

QB3. To what extent do you agree or disagree that your employer provides you with the necessary tools or training to work effectively with the most recent digital technologies, including Artificial Intelligence? (%)



[30] QB3. To what extent do you agree or disagree that your employer provides you with the necessary tools or training to work effectively with the most recent digital technologies, including Artificial Intelligence? Totally agree; Tend to agree; Tend to disagree; Totally disagree; Not applicable (SPONTANEOUS); Don't know.



Conclusion

The results of this Special Eurobarometer illustrate that, overall, Europeans hold a positive view of the impact of the most recent digital technologies, including Artificial Intelligence, even though some positive perceptions have declined over the past seven years. Almost two thirds of the respondents in employment have a positive opinion of these technologies for their job. Furthermore, of all respondents, six in ten believe they have a positive impact on the economy, on quality of life and on social security benefits, and at least half say this of their impact on society. Nonetheless, the shares of those who have a positive opinion have declined since March 2017, especially about the impact of these technologies on the economy.

In line with these findings, while more than six in ten think that robots and Artificial Intelligence are a good thing for society, this proportion has decreased by five percentage points since March 2017. In addition, a very large majority think that these technologies require careful management and less than half would like to see them used more widely outside the workplace. Finally, two thirds are concerned that, due to the use of robots and Artificial Intelligence, more jobs will disappear than new jobs will be created, and state that Artificial Intelligence steal people's jobs – even though both these proportions have declined since 2017.

EU citizens also tend to consider their level of digital skills as sufficient in their daily life, to be able to benefit from digital and online learning opportunities, and perform their current or a future job. This result should be interpreted considering the significant number of Europeans who lack even basic digital skills (almost 45%)[31] and the growing but still relatively low number of ICT specialists (est. 9.8 million)[32]. There are country-level variations in perceived digital literacy, with respondents in Denmark, Finland, Luxembourg, the Netherlands and Sweden tending to be more confident about their level of digital skills than those in Greece, Hungary, Portugal and Romania. On the bright side, among those currently employed, more than two thirds indicate that their employer provides them with the necessary tools or training to work effectively with the most recent digital technologies.

Results are mixed when it comes to perceptions of digital technologies and Artificial Intelligence in the workplace. Large majorities positively perceive the use of these technologies in the workplace, and agree that they can do boring or repetitive jobs. There is also agreement that they can increase the pace at which workers complete tasks, a positive effect that should however be seen in light of possible work intensification, which creates both individual and societal risks. The majority believes that these technologies can be used to make accurate decisions in the workplace. Furthermore, the use of digital technologies to improve workers' safety and security is generally perceived as positive. A relative majority holds a positive view of their use to allocate tasks to workers or manage their working schedules and shifts.

However, around six in ten agree that robots and Artificial Intelligence have a negative impact on communication between colleagues. Less than half of respondents have a favourable opinion of the use of these technologies to perform other activities in the workplace: only a minority positively sees their use to collect, process, and store workers' personal data, to gather additional information on applicants for a job, to select applicants for a job, to assess workers' performance, to monitor workers, or to automatically fire workers.

It is therefore unsurprising that large majorities of EU citizens consider that a series of measures to address risks and maximising the benefits of digital technologies would be important: protecting workers' privacy; involving workers and their representatives in the design and adoption of new technologies; enforcing more transparency in the use of digital technologies to handle HR decision-making; prohibiting fully automated decision-making processes; and limiting the automated monitoring of employees.

In terms of awareness and level of information about the use of these technologies in the workplace, most employees are aware of the use their employer makes of digital technologies, including Artificial Intelligence, to manage their and their co-workers' activities. Sizeable proportions of those who have a current or past occupation also indicate that digital technologies have performed activities in their workplace, and this is especially the case for enforcing safety measures and monitoring workers' activities. Moreover, around half of those currently in employment say their employer informed them about the use of digital technologies to manage activities in their workplace, while slightly more than half among employers and managers report having done so.

Finally, there are noticeable patterns in the results of this survey according to the socio-demographic characteristics of the respondents. In particular, younger respondents, those with a higher education level, those with the least financial difficulties, and those who are currently working are the most likely to hold positive attitudes towards digital technologies and Artificial Intelligence (including with regard

[31] 'Digital skills in 2023: impact of education and age': https://ec.europa.eu/eurostat/web/products-eurostat-news/w/ddn-20240222-1

[32] 'More people employed in ICT in the EU in 2023': https://ec.europa.eu/eurostat/web/products-eurostat-news/w/ddn-20240524-2

to their impact on employment and the labour market) and towards their use in the workplace. In addition, there are differences between sectors and size of the establishments when it comes to reported uptake of digital technologies and information about their use in the workplace, with these figures being highest in the logistics sector as well as in larger establishments.

Overall, these findings underscore the nuanced views that Europeans hold towards digital technologies and Artificial Intelligence, reflecting a balance of optimism about their benefits and concerns about their potential risks and implications for the future of work.

Technical Specifications

Between 25 April and 22 May 2024, Verian (former Kantar Public) on behalf of Verian Belgium carried out the wave 101.4 of the Eurobarometer survey, on request of the European Commission, Directorate-General for Communication, "Media monitoring and Eurobarometer" Unit.

Wave 101.4 covers the population of the respective nationalities of the European Union Member States, resident in each of the 27 Member States and aged 15 years and over.

The basic sample design applied in all countries is a stratified multi-stage, random (probability) one. In each country, the sample frame is first stratified by NUTS regions and within each region by a measure of urbanity (DEGURBA). The number of sample points selected in each strata reflects the stratum population 15+. At the second stage sampling points were drawn with probability proportional to their 0+ population size from within each stratum. The samples thus represent the whole territory of the countries surveyed according to the EUROSTAT NUTS II (or equivalent) and according to the distribution of the resident population of the respective nationalities in terms of metropolitan, urban and rural areas33.

In each of the selected sampling points, a starting coordinate was drawn at random and a reverse geo-coding tool used to identify the closest address to the coordinate. This address was the starting address for the random walk. Further addresses (every Nth address) were selected by standard "random route" procedures, from the initial address. In each household, the respondent was drawn, at random. The approach to the random selection was conditional on the household size. By way of example for households with two 15+ members the script was used to select either the informant (person responding to the screener questionnaire) or the other eligible member in the household. For households with three 15+ members the script was used to select either the informant (1/3 of the time) or the two other eligible members in the household (2/3 of the time). Where the two other members were selected, the interviewer was then told to either ask for the youngest or oldest. The script would randomly assign the selection to youngest or oldest with equal probability. This process continues for four 15+ household members – randomly asking for the youngest, 2nd youngest and oldest.

For households with five 15+ members we revert to the last birthday rule.

If no contact was made with anyone in the household, or if the respondent selected was not available (busy), the interviewer revisited the same household up to three additional times (four contact attempts in total). Interviewers never indicate that the survey is conducted on behalf of the European Commission beforehand; they may give this information once the survey is completed, upon request. The recruitment phase was slightly different in the Netherlands, Finland, and Sweden. In the two latter countries, a sample of addresses within each sampling point were selected from the address or population register (in Finland, selection is not done in all sample points, but in some where response rates are expected to improve). The selection of addresses was done in a random manner. Households were then contacted by telephone and recruited to take part in the survey. In the Netherlands, a dual frame RDD sample (mobile and landline numbers) are used as there is no comprehensive population register with telephone numbers available. The selection of numbers on both frames is done in a random manner with each number getting an equal probability of selection. Unlike Sweden and Finland, the sample is un-clustered.

[33] Urban Rural classification based on DEGURBA (https://ec.europa.eu/eurostat/web/degree-of-urbanisation/background)

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	COUNTRIES	INSTITUTES	N° INTERVIEWS	FIELDWORK DATES		POPULATION 15+	PROPORTION EU27
BE	Belgium	MCM Belgium	1,019	26-04-2024	15-05-2024	9,801,547	2.6%
BG	Bulgaria	Kantar TNS BBSS	1,018	25-04-2024	19-05-2024	5,533,938	1.4%
CZ	Czechia	STEM/MARK	1,019	29-04-2024	12-05-2024	9,075,934	2.4%
DK	Denmark	Mantle Denmark (Verian)	1,003	27-04-2024	22-05-2024	4,984,048	1.3%
DE	Germany	Mantle Germany (Verian)	1,603	29-04-2024	21-05-2024	72,405,020	19.0%
EE	Estonia	Norstat Eesti	1,002	26-04-2024	16-05-2024	1,141,759	0.3%
IE	Ireland	B and A Research	1,006	26-04-2024	16-05-2024	4,250,998	1.1%
EL	Greece	Kantar Greece	1,007	25-04-2024	18-05-2024	9,019,518	2.4%
ES	Spain	Mantle Spain (Verian)	1,000	30-04-2024	20-05-2024	41,533,486	10.9%
FR	France	MCM France	1,002	26-04-2024	16-05-2024	56,365,353	14.8%
HR	Croatia	Hendal	1,005	26-04-2024	14-05-2024	3,301,831	0.9%
IT	Italy	Testpoint Italia	1,037	29-04-2024	10-05-2024	51,632,657	13.5%
CY	Rep. Of Cyprus	CYMAR Market Research	504	25-04-2024	16-05-2024	772,320	0.2%
LV	Latvia	Kantar TNS Latvia	1,010	26-04-2024	17-05-2024	1,582,326	0.4%
LT	Lithuania	Norstat LT	1,014	26-04-2024	14-05-2024	2,429,823	0.6%
LU	Luxembourg	ILRES	507	26-04-2024	16-05-2024	555,900	0.1%
HU	Hungary	Kantar Hoffmann	1,012	29-04-2024	17-05-2024	8,205,783	2.1%
MT	Malta	MISCO International	500	29-04-2024	21-05-2024	473,015	0.1%
NL	Netherlands	MCM Netherlands	1,011	30-04-2024	15-05-2024	15,081,342	4.0%
AT	Austria	Das Österreichische Gallup Ins.	1,007	29-04-2024	14-05-2024	7,788,036	2.0%
PL	Poland	Research Collective	1,007	26-04-2024	15-05-2024	31,079,533	8.1%
PT	Portugal	Intercampus SA	1,029	01-05-2024	17-05-2024	9,113,419	2.4%
RO	Romania	CSOP SRL	1,042	26-04-2024	15-05-2024	15,981,575	4.2%
SI	Slovenia	Mediana DOO	1,003	25-04-2024	15-05-2024	1,799,078	0.5%
SK	Slovakia	MNFORCE	1,023	27-04-2024	14-05-2024	4,554,569	1.2%
FI	Finland	Taloustutkimus Oy	1,006	25-04-2024	16-05-2024	4,722,540	1.2%
SE	Sweden	Mantle Sweden (Verian)	1,019	29-04-2024	16-05-2024	8,541,497	2.2%
		TOTAL EU27	26,415	25-04-2024	22-05-2024	381,726,845	100%

* It should be noted that the total percentage shown in this table may exceed 100% due to rounding.

Interviewing mode per country

Interviews were conducted through face-to-face interviews, either physically in people's homes or through remote video interaction in the appropriate national language. Interviews with remote video interaction ("online face-to-face" or CAVI, Computer Assisted Video Interviewing, were conducted only in Czechia, Denmark, Germany, Finland and Malta).

	COUNTRIES	N° OF CAPI INTERVIEWS	N° OF CAVI INTERVIEWS	TOTAL N° INTERVIEWS
BE	Belgium	1,019		1,019
BG	Bulgaria	1,018		1,018
CZ	Czechia	820	199	1,019
DK	Denmark	700	303	1,003
DE	Germany	1,475	128	1,603
EE	Estonia	1,002		1,002
IE	Ireland	1,006		1,006
EL	Greece	1,007		1,007
ES	Spain	1,000		1,000
FR	France	1,002		1,002
HR	Croatia	1,005		1,005
IT	Italy	1,037		1,037
CY	Rep. Of Cyprus	504		504
LV	Latvia	1,010		1,010
LT	Lithuania	1,014		1,014
LU	Luxembourg	507		507
HU	Hungary	1,012		1,012
MT	Malta	349	151	500
NL	Netherlands	1,011		1,011
AT	Austria	1,007		1,007
PL	Poland	1,007		1,007
PT	Portugal	1,029		1,029
RO	Romania	1,042		1,042
SI	Slovenia	1,003		1,003
SK	Slovakia	1,023		1,023
FI	Finland	708	298	1,006
SE	Sweden	1,019		1,019
	TOTAL EU27	25,336	1,079	26,415

CAPI : Computer-Assisted Personal interviewing
CAVI : Computer-Assisted Video interviewing

Response rates

For each country a comparison between the responding sample and the universe (i.e. the overall population in the country) is carried out. Weights are used to match the responding sample to the universe on gender by age, region and degree of urbanisation. For European estimates (i.e. EU average), an adjustment is made to the individual country weights, weighting them up or down to reflect their 15+ population as a proportion of the EU 15+ population.

The response rates are calculated by dividing the total number of complete interviews with the number of all the addresses visited, apart from ones that are not eligible but including those where eligibility is unknown. For wave 101.4 of the EUROBAROMETER survey, the response rates for the EU27 countries, calculated by Verian (former Kantar Public), are:

	COUNTRIES	RESPONSE RATES
BE	Belgium	61.3%
BG	Bulgaria	48.3%
CZ	Czechia	51.4%
DK	Denmark	30.9%
DE	Germany	31.9%
EE	Estonia	78.9%
IE	Ireland	36.5%
EL	Greece	30.7%
ES	Spain	31.3%
FR	France	41.1%
HR	Croatia	45.9%
IT	Italy	28.5%
CY	Rep. Of Cyprus	63.5%
LV	Latvia	37.8%
LT	Lithuania	47.9%
LU	Luxembourg	29.1%
HU	Hungary	58.2%
MT	Malta	60.2%
NL	Netherlands	81.6%
AT	Austria	41.9%
PL	Poland	46.1%
PT	Portugal	49.4%
RO	Romania	59.8%
SI	Slovenia	41.0%
SK	Slovakia	53.6%
FI	Finland	35.9%
SE	Sweden	80.6%

Margins of error

Readers are reminded that survey results are estimations, the accuracy of which, everything being equal, rests upon the sample size and upon the observed percentage. With samples of about 1,000 interviews, the real percentages vary within the following confidence limits:

Statistical Margins due to the sampling process

(at the 95% level of confidence)

various sample sizes are in rows *various observed results are in columns*

	5% 95%	10% 90%	15% 85%	20% 80%	25% 75%	30% 70%	35% 65%	40% 60%	45% 55%	50% 50%	
N=50	6,0	8,3	9,9	11,1	12,0	12,7	13,2	13,6	13,8	13,9	N=50
N=500	1,9	2,6	3,1	3,5	3,8	4,0	4,2	4,3	4,4	4,4	N=500
N=1000	1,4	1,9	2,2	2,5	2,7	2,8	3,0	3,0	3,1	3,1	N=1000
N=1500	1,1	1,5	1,8	2,0	2,2	2,3	2,4	2,5	2,5	2,5	N=1500
N=2000	1,0	1,3	1,6	1,8	1,9	2,0	2,1	2,1	2,2	2,2	N=2000
N=3000	0,8	1,1	1,3	1,4	1,5	1,6	1,7	1,8	1,8	1,8	N=3000
N=4000	0,7	0,9	1,1	1,2	1,3	1,4	1,5	1,5	1,5	1,5	N=4000
N=5000	0,6	0,8	1,0	1,1	1,2	1,3	1,3	1,4	1,4	1,4	N=5000
N=6000	0,6	0,8	0,9	1,0	1,1	1,2	1,2	1,2	1,3	1,3	N=6000
N=7000	0,5	0,7	0,8	0,9	1,0	1,1	1,1	1,1	1,2	1,2	N=7000
N=7500	0,5	0,7	0,8	0,9	1,0	1,0	1,1	1,1	1,1	1,1	N=7500
N=8000	0,5	0,7	0,8	0,9	0,9	1,0	1,0	1,1	1,1	1,1	N=8000
N=9000	0,5	0,6	0,7	0,8	0,9	0,9	1,0	1,0	1,0	1,0	N=9000
N=10000	0,4	0,6	0,7	0,8	0,8	0,9	0,9	1,0	1,0	1,0	N=10000
N=11000	0,4	0,6	0,7	0,7	0,8	0,9	0,9	0,9	0,9	0,9	N=11000
N=12000	0,4	0,5	0,6	0,7	0,8	0,8	0,9	0,9	0,9	0,9	N=12000
N=13000	0,4	0,5	0,6	0,7	0,7	0,8	0,8	0,8	0,9	0,9	N=13000
N=14000	0,4	0,5	0,6	0,7	0,7	0,8	0,8	0,8	0,8	0,8	N=14000
N=15000	0,3	0,5	0,6	0,6	0,7	0,7	0,8	0,8	0,8	0,8	N=15000
	5% 95%	10% 90%	15% 85%	20% 80%	25% 75%	30% 70%	35% 65%	40% 60%	45% 55%	50% 50%	

Questionnaire

Introduction: Digital technologies refer to technologies that use data and computing to help people do their work more efficiently. They include hardware such as smartphones, laptops or robots, and software, such as smartphone applications, video conferencing tools, collaboration platforms, or project management tools in the workplace.
Artificial Intelligence refers to computer systems capable of performing tasks that typically require human intelligence. It is used, for instance, in driverless cars or drones, in healthcare to improve medical diagnoses, and in various other applications such as answering questions and offering support to users on websites or call centres. Artificial Intelligence can also be used in workplaces, for example to monitor and evaluate workers or automatically assign them certain tasks. Artificial Intelligence can also be used to create new content such as images, text, or music as well as to answer questions and assist with tasks such as composing emails or essays. ChatGPT is the best-known example.

QB1. In your view, what impact do the most recent digital technologies, including Artificial Intelligence, currently have on:

(ALL RESPONDENTS except item 4: RESPONDENTS WHO HAVE A CURRENT OCCUPATION - READ OUT - ROTATE - ONE ANSWER ONLY)

	A very positive impact	A fairly positive impact	A fairly negative impact	A very negative impact	It depends [SPONTEANOUS]	Do not know enough about the most recent digital technologies [SPONTEANOUS]	Do not know [SPONTEANOUS]
The economy	1	2	3	4	5	6	7
Society	1	2	3	4	5	6	7
Your quality of life	1	2	3	4	5	6	7
Your current job (N)	1	2	3	4	5	6	7
Social security benefits (N)	1	2	3	4	5	6	7

Modified trend EB87.1 EBS 460-2017, except statements 4 and 5 – 2.5 QU

QB2. To what extent do you agree or disagree with the following statements regarding your skills in the use of the most recent digital technologies, including Artificial Intelligence? Think about, for example, using smart appliances at home that you can control remotely for heating, cooking, or cleaning; using apps on your smartphone to handle your daily schedule; or using websites to manage your tasks at work.
You consider yourself to be sufficiently skilled in the use of digital technologies... (M)

(ALL RESPONDENTS - READ OUT ROTATE - ONE ANSWER ONLY)

	Totally agree	Tend to agree	Tend to disagree	Totally disagree	Not applicable (N)	Do not know [SPONTEANOUS]
... in your daily life	1	2	3	4	5	6
... to do your job	1	2	3	4	5	6
... to do a future job if you were to find a job or to change jobs within the next twelve months	1	2	3	4	5	6
... to be able to benefit from digital and online learning opportunities (M)	1	2	3	4	5	6

Modified trend QD4 EB87.1 of EBS 460-2017 – 2 QU

QB3. To what extent do you agree or disagree that your employer provides you with the necessary tools or training to work effectively with the most recent digital technologies, including Artificial Intelligence?

(RESPONDENTS WHO HAVE A CURRENT OCCUPATION - READ OUT, ONE ANSWER ONLY)

1. Totally agree
2. Tend to agree
3. Tend to disagree
4. Totally disagree
5. Not applicable [SPONTEANOUS]
6. Do not know [SPONTEANOUS]

New – 1 QU

QB4. Overall, how aware are you of the use that your employer makes of digital technologies, including Artificial Intelligence, to manage your or your co-**workers' activities? Examples include setting working** schedules, monitoring your activities, or evaluating performance.

(RESPONDENTS WHO WORK FOR AN EMPLOYER OR MANAGER - READ OUT, ONE ANSWER ONLY)

1. Totally aware
2. Fairly aware
3. Fairly unaware
4. Totally unaware
5. Your workplace does not use digital technologies [SPONTANEOUS]
6. Do not know [SPONTEANOUS]

New – 1 QU

QB5. How positively or negatively do you perceive the use of robots and Artificial Intelligence in the workplace?

By robots we mean machines designed to perform tasks which are normally performed by humans. They can perform them by themselves or, often, working together with a human. For example, robots that can perform surgery together with a surgeon or automatic tills in supermarkets.

(ALL RESPONDENTS - READ OUT ROTATE - ONE ANSWER ONLY)

1. Very positively
2. Fairly positively
3. Fairly negatively
4. Very negatively
5. Do not know [SPONTEANOUS]

Modified trend on QD10 EB87.1 of EBS 460-2017 – 1 QU

QB6. To what extent do you agree or disagree with each of the following statements?

(ALL RESPONDENTS - READ OUT ROTATE - ONE ANSWER ONLY)

	Totally agree	Tend to agree	Tend to disagree	Totally disagree	Do not know [SPONTEANOUS]
Due to the use of robots and artificial intelligence, more jobs will disappear than new jobs will be created	1	2	3	4	5
Robots and Artificial Intelligence are a good thing for society, because they help people do their jobs or carry out daily tasks at home	1	2	3	4	5
Robots and Artificial Intelligence are technologies that require careful management	1	2	3	4	5
Artificial Intelligence is necessary as it can do jobs that are seen as boring or repetitive (N)	1	2	3	4	5
Robots and Artificial Intelligence steal people's jobs	1	2	3	4	5
Robots and Artificial Intelligence increase the pace at which workers complete tasks (N)	1	2	3	4	5
Robots and Artificial Intelligence have a negative impact on communication between colleagues (N)	1	2	3	4	5
Robots and Artificial Intelligence can be used to make accurate decisions in the workplace (N)	1	2	3	4	5
Robots and Artificial Intelligence should be used more widely outside the workplace (N)	1	2	3	4	5

Modified trend QD12 EB87.1 of EBS 460-2017, except items 4, 6, 7, 8, 9 – 4.5 QU

QB7. Have the following activities ever been performed by digital technologies, including Artificial Intelligence, in your current or previous workplaces? For each activity, please refer to your most recent workplace experience.

(RESPONDENTS WHO HAVE A CURRENT OR PAST OCCUPATION - READ OUT – ROTATE - ONE ANSWER ONLY)

	Yes, all the time	Yes, often	No, rarely	No, never	Not applicable [SPONTANEOUS]	Do not know [SPONTEANOUS]
Hiring workers	1	2	3	4	5	6
Allocating tasks to workers	1	2	3	4	5	6
Managing worktime schedules	1	2	3	4	5	6
Monitoring workers' activities	1	2	3	4	5	6
Assessing workers' performance, including imposing sanctions or attributing rewards	1	2	3	4	5	6
Enforcing safety measures	1	2	3	4	5	6

New – 3 QU

QB8. In general, how positively or negatively do you perceive the use of digital technologies, including Artificial Intelligence, in the workplace for each of the following activities:

(ALL RESPONDENTS - READ OUT ROTATE - ONE ANSWER ONLY)

	Very positively	Somewhat positively	Somewhat negatively	Very negatively	Do not know [SPONTEANOUS]
Gathering additional information on applicants for a job	1	2	3	4	5
Selecting applicants for a job	1	2	3	4	5
Allocating tasks to workers or managing their working schedules and shifts	1	2	3	4	5
Collecting, processing, and storing workers' personal data	1	2	3	4	5
Improving workers' safety and security	1	2	3	4	5
Monitoring workers	1	2	3	4	5
Assessing workers' performance	1	2	3	4	5
Automatically firing workers	1	2	3	4	5

New – 4 QU

QB9. Has your employer informed you about the use of digital technologies, including Artificial Intelligence, to manage activities in your workplace?

(RESPONDENTS WHO WORK FOR AN EMPLOYER OR MANAGER - READ OUT, ONE ANSWER ONLY except for items 2,3 and 4)

1. Yes, you have been made aware of the use of these technologies, but without further details
2. Yes, and you have received a detailed explanation, including information about the benefits, drawbacks, and your rights
3. Yes, and you have been given access to your personal data
4. Yes, and you have been given access to the results of the automated analysis carried out
5. No, you have not been made aware
6. Not applicable [SPONTEANOUS]
7. Don't know [SPONTEANOUS]

New – 1 QU

QB10. Have you ensured that your employees or the people you manage have been informed about the use of digital technologies, including Artificial Intelligence, to manage activities in your workplace?

(RESPONDENTS WHO ARE EMPLOYERS OR MANAGERS - READ OUT, ONE ANSWER ONLY except for items 2, 3 and 4)

1. Yes, you have made them aware of it, but without further details
2. Yes, and you have given them a detailed explanation, including information about benefits, drawbacks, and their rights
3. Yes, and you have given them access to their personal data
4. Yes, and you have given them the results of any automated analysis that concerns them
5. No, you have not made them aware of it
6. Don't know [SPONTEANOUS]

New – 1 QU

QB11. How important, if at all, do you think the following rules would be in addressing risks and maximizing the benefits of digital technologies, including Artificial Intelligence, in the workplace?

(ALL RESPONDENTS - READ OUT - ROTATE – ONE ANSWER ONLY) –

	Very important	Somewhat important	Not very important	Not at all important	Do not know [SPONTEANOUS]
Protecting workers' privacy	1	2	3	4	5
Prohibiting fully automated decision-making processes	1	2	3	4	5
Limiting the automated monitoring of employees	1	2	3	4	5
Enforcing more transparency in the use of digital technologies to handle HR decision-making	1	2	3	4	5
Involving workers and their representatives in the design and adoption of new technologies	1	2	3	4	5

New – 2.5 QU



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